

8

Hankel Norm Approximation

This chapter is devoted to the study of optimal Hankel norm approximation and its applications in $\mathcal{L}_\infty$ norm model reduction. The Hankel operator is introduced first together with some time domain and frequency domain characterizations. The optimal Hankel norm approximation problem can be stated as follows: Given $G(s)$ of McMillan degree n , find $\hat{G}(s)$ of McMillan degree $k < n$ such that $\|G(s) - \hat{G}(s)\|_H$ is minimized. The solution to this approximation problem relies on the all-pass dilation result of a square transfer function which will be given for a general class of transfer functions. The all-pass dilation results are then specialized to obtain the optimal Hankel norm approximations, which gives

$$\inf \|G(s) - \hat{G}(s)\|_H = \sigma_{k+1}$$

where $\sigma_1 > \sigma_2 \dots > \sigma_{k+1} \dots > \sigma_n$ are the Hankel singular values of $G(s)$. Moreover, we show that a square stable transfer function $G(s)$ can be represented as

$$G(s) = D_0 + \sigma_1 E_1(s) + \sigma_2 E_2(s) + \dots + \sigma_n E_n(s)$$

where $E_k(s)$ are all-pass functions and the partial sum $D_0 + \sigma_1 E_1(s) + \sigma_2 E_2(s) + \dots + \sigma_k E_k(s)$ have McMillan degrees k . This representation is obtained by reducing the order one dimension at a time via optimal Hankel norm approximations. This representation also gives that

$$\|G\|_\infty \leq 2(\sigma_1 + \dots + \sigma_n)$$

and further that there exists a constant D_0 such that

$$\|G(s) - D_0\|_\infty \leq (\sigma_1 + \dots + \sigma_n).$$

The above bounds are then used to show that the k -th order optimal Hankel norm approximation, $\hat{G}(s)$, together with some constant matrix D_0 satisfies

$$\left\| G(s) - \hat{G}(s) - D_0 \right\|_{\infty} \leq (\sigma_{k+1} + \dots + \sigma_n).$$

We shall also provide an alternative proof for the error bounds derived in the last chapter for the truncated balanced realizations using the results obtained in this chapter.

Finally we consider the Hankel operator in discrete time and offer some alternative proof of the well-known Nehari's theorem.

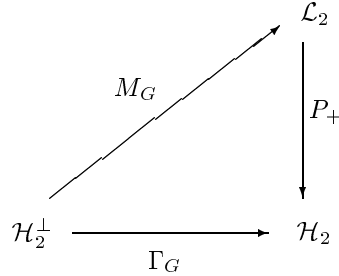
8.1 Hankel Operator

Let $G(s) \in \mathcal{L}_{\infty}$ be a matrix function. The *Hankel operator* associated with G will be denoted by Γ_G and is defined as

$$\Gamma_G : \mathcal{H}_2^{\perp} \mapsto \mathcal{H}_2$$

$$\Gamma_G f := (P_+ M_G) f = P_+(Gf), \quad \text{for } f \in \mathcal{H}_2^{\perp}$$

i.e., $\Gamma_G = P_+ M_G|_{\mathcal{H}_2^{\perp}}$. This is shown in the following diagram:



There is a corresponding Hankel operator in the time domain. Let $g(t)$ denote the inverse (bilateral) Laplace transform of $G(s)$. Then the time domain Hankel operator is

$$\Gamma_g : \mathcal{L}_2(-\infty, 0] \mapsto \mathcal{L}_2[0, \infty)$$

$$\Gamma_g f := P_+(g * f), \quad \text{for } f \in \mathcal{L}_2(-\infty, 0].$$

Thus

$$(\Gamma_g f)(t) = \begin{cases} \int_{-\infty}^0 g(t - \tau) f(\tau) d\tau, & t \geq 0; \\ 0, & t < 0. \end{cases}$$

Because of the isometric isomorphism property between the $\mathcal{L}_2$ spaces in the time domain and in the frequency domain, we have

$$\|\Gamma_g\| = \|\Gamma_G\|.$$

Hence, in this book we will use the time domain and the frequency domain descriptions for Hankel operators interchangeably.

For the interest of this book, we will now further restrict G to be rational, i.e., $G(s) \in \mathcal{RL}_\infty$. Then G can be decomposed into strictly causal part and anticausal part, i.e., there are $G_s(s) \in \mathcal{RH}_2$ and $G_u(s) \in \mathcal{RH}_2^\perp$ such that

$$G(s) = G_s(s) + G(\infty) + G_u(s).$$

Now for any $f \in \mathcal{H}_2^\perp$, it is easy to see that

$$\Gamma_G f = P_+(Gf) = P_+(G_s f).$$

Hence, the Hankel operator associated with $G \in \mathcal{RL}_\infty$ depends only on the strictly causal part of G . In particular, if G is antistable, i.e., $G^\sim(s) \in \mathcal{RH}_\infty$, then $\Gamma_G = 0$. Therefore, there is no loss of generality in assuming $G \in \mathcal{RH}_\infty$ and strictly proper.

The adjoint operator of Γ_G can be computed easily from the definition as below: let $f \in \mathcal{H}_2^\perp, g \in \mathcal{H}_2$, then

$$\begin{aligned} \langle \Gamma_G f, g \rangle &:= \langle P_+ G f, g \rangle \\ &= \langle P_+ G f, g \rangle + \langle P_- G f, g \rangle \quad (\text{since } P_- G f \text{ and } g \text{ are orthogonal}) \\ &= \langle G f, g \rangle \\ &= \langle f, G^\sim g \rangle \\ &= \langle f, P_+ G^\sim g \rangle + \langle f, P_- G^\sim g \rangle \\ &= \langle f, P_- G^\sim g \rangle \quad (\text{since } f \text{ and } P_+ G^\sim g \text{ are orthogonal}). \end{aligned}$$

Hence $\Gamma_G^* g = P_-(G^\sim g) : \mathcal{H}_2 \mapsto \mathcal{H}_2^\perp$ or $\Gamma_G^* = P_- M_{G^\sim}|_{\mathcal{H}_2}$.

Now suppose $G \in \mathcal{RH}_\infty$ has a state space realization as given below:

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned}$$

with A stable and $x(-\infty) = 0$. Then the Hankel operator Γ_g can be written as

$$\Gamma_g u(t) = \int_{-\infty}^0 C e^{A(t-\tau)} B u(\tau) d\tau, \quad \text{for } t \geq 0$$

and has the interpretation of the system future output

$$y(t) = \Gamma_g u(t), \quad t \geq 0$$

based on the past input $u(t), t \leq 0$.

In the state space representation, the Hankel operator can be more specifically decomposed as the composition of maps from the past input to the initial state and then

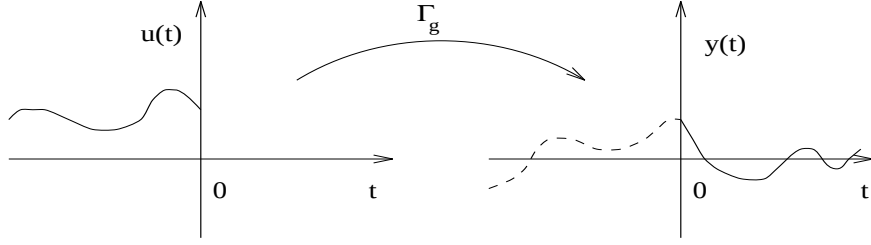


Figure 8.1: System theoretical interpretation of Hankel operators

from the initial state to the future output. These two operators will be called the *controllability operator*, Ψ_c , and the *observability operator*, Ψ_o , respectively, and are defined as

$$\begin{aligned}\Psi_c &: \mathcal{L}_2(-\infty, 0] \mapsto \mathbb{C}^n \\ \Psi_c u &:= \int_{-\infty}^0 e^{-A\tau} B u(\tau) d\tau\end{aligned}$$

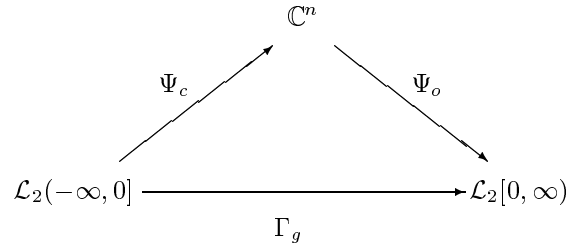
and

$$\begin{aligned}\Psi_o &: \mathbb{C}^n \mapsto \mathcal{L}_2[0, \infty) \\ \Psi_o x_0 &:= C e^{At} x_0, \quad t \geq 0.\end{aligned}$$

(If all the data are real, then the two operators become $\Psi_c : \mathcal{L}_2(-\infty, 0] \mapsto \mathbb{R}^n$ and $\Psi_o : \mathbb{R}^n \mapsto \mathcal{L}_2[0, \infty)$.) Clearly, $x_0 = \Psi_c u(t)$ for $u(t) \in \mathcal{L}_2(-\infty, 0]$ is the system state at $t = 0$ due to the past input and $y(t) = \Psi_o x_0$, $t \geq 0$, is the future output due to the initial state x_0 with the input set to zero.

It is easy to verify that

$$\Gamma_g = \Psi_o \Psi_c.$$



The adjoint operators of Ψ_c and Ψ_o can also be obtained easily from their definitions as follows: let $u(t) \in \mathcal{L}_2(-\infty, 0]$, $x_0 \in \mathbb{C}^n$, and $y(t) \in \mathcal{L}_2[0, \infty)$, then

$$\langle \Psi_c u, x_0 \rangle_{\mathbb{C}^n} = \int_{-\infty}^0 u^*(\tau) B^* e^{-A^* \tau} x_0 d\tau = \langle u, B^* e^{-A^* \tau} x_0 \rangle_{\mathcal{L}_2(-\infty, 0]} = \langle u, \Psi_c^* x_0 \rangle_{\mathcal{L}_2(-\infty, 0]}$$

and

$$\langle \Psi_o x_0, y \rangle_{\mathcal{L}_2[0, \infty)} = \int_0^\infty x^*(t) e^{A^*t} C^* y(t) dt = \langle x_0, \int_0^\infty e^{A^*t} C^* y(t) dt \rangle_{\mathbb{C}^n} = \langle x_0, \Psi_o^* y \rangle_{\mathbb{C}^n}$$

where $\langle \cdot, \cdot \rangle_X$ denotes the inner product in the Hilbert space X . Therefore, we have

$$\begin{aligned} \Psi_c^* : \mathbb{C}^n &\mapsto \mathcal{L}_2(-\infty, 0] \\ \Psi_c^* x_0 &= B^* e^{-A^* \tau} x_0, \quad \tau \leq 0 \end{aligned}$$

and

$$\begin{aligned} \Psi_o^* : \mathcal{L}_2[0, \infty) &\mapsto \mathbb{C}^n \\ \Psi_o^* y(t) &= \int_0^\infty e^{A^*t} C^* y(t) dt. \end{aligned}$$

This also gives the adjoint of Γ_g :

$$\begin{aligned} \Gamma_g^* &= (\Psi_o \Psi_c)^* = \Psi_c^* \Psi_o^* : \mathcal{L}_2[0, \infty) \mapsto \mathcal{L}_2(-\infty, 0] \\ \Gamma_g^* y &= \int_0^\infty B^* e^{A^*(t-\tau)} C^* y(t) dt, \quad \tau \leq 0. \end{aligned}$$

Let L_c and L_o be the controllability and observability Gramians of the system, i.e.,

$$\begin{aligned} L_c &= \int_0^\infty e^{At} B B^* e^{A^*t} dt \\ L_o &= \int_0^\infty e^{A^*t} C^* C e^{At} dt. \end{aligned}$$

Then we have

$$\begin{aligned} \Psi_c \Psi_c^* x_0 &= L_c x_0 \\ \Psi_o^* \Psi_o x_0 &= L_o x_0 \end{aligned}$$

for every $x_0 \in \mathbb{C}^n$. Thus L_c and L_o are the matrix representations of the operators $\Psi_c \Psi_c^*$ and $\Psi_o^* \Psi_o$.

Theorem 8.1 *The operator $\Gamma_g^* \Gamma_g$ (or $\Gamma_G^* \Gamma_G$) and the matrix $L_c L_o$ have the same nonzero eigenvalues. In particular $\|\Gamma_g\| = \sqrt{\rho(L_c L_o)}$.*

Proof. Let $\sigma^2 \neq 0$ be an eigenvalue of $\Gamma_g^* \Gamma_g$, and let $0 \neq u \in \mathcal{L}_2(-\infty, 0]$ be a corresponding eigenvector. Then by definition

$$\Gamma_g^* \Gamma_g u = \Psi_c^* \Psi_o^* \Psi_o \Psi_c u = \sigma^2 u. \quad (8.1)$$

Pre-multiply (8.1) by Ψ_c and define $x = \Psi_c u \in \mathbb{C}^n$ to get

$$L_c L_o x = \sigma^2 x. \quad (8.2)$$

Note that $x = \Psi_c u \neq 0$ since otherwise $\sigma^2 u = 0$ from (8.1) which is impossible. So σ^2 is an eigenvalue of $L_c L_o$.

On the other hand, suppose $\sigma^2 \neq 0$ and $x \neq 0$ are an eigenvalue and a corresponding eigenvector of $L_c L_o$. Pre-multiply (8.2) by $\Psi_c^* L_o$ and define $u = \Psi_c^* L_o x$ to get (8.1). It is easy to see that $u \neq 0$ since $\Psi_c u = \Psi_c \Psi_c^* L_o x = L_c L_o x = \sigma^2 x \neq 0$. Therefore σ^2 is an eigenvalue of $\Gamma_g^* \Gamma_g$.

Finally, since $G(s)$ is rational, $\Gamma_g^* \Gamma_g$ is compact and self-adjoint and has only discrete spectrum. Hence $\|\Gamma_g\|^2 = \|\Gamma_g^* \Gamma_g\| = \rho(L_c L_o)$. $\square$

Remark 8.1 Let $\sigma^2 \neq 0$ be an eigenvalue of $\Gamma_g^* \Gamma_g$ and $0 \neq u \in \mathcal{L}_2(-\infty, 0]$ be a corresponding eigenvector. Define

$$v := \frac{1}{\sigma} \Gamma_g u \in \mathcal{L}_2[0, \infty).$$

Then (u, v) satisfy

$$\begin{aligned} \Gamma_g u &= \sigma v \\ \Gamma_g^* v &= \sigma u. \end{aligned}$$

This pair of vectors (u, v) are called a *Schmidt pair* of Γ_g . The proof given above suggests a way to construct this pair: find the eigenvalues and eigenvectors of $L_c L_o$, i.e., σ_i^2 and x_i such that

$$L_c L_o x_i = \sigma_i^2 x_i.$$

Then the pairs (u_i, v_i) given below are the corresponding Schmidt pairs:

$$u_i = \Psi_c^* \left(\frac{1}{\sigma_i} L_o x_i \right) \in \mathcal{L}_2(-\infty, 0], \quad v_i = \Psi_o x_i \in \mathcal{L}_2[0, \infty).$$

♡

Remark 8.2 As seen in various literature, there are some alternative ways to write a Hankel operator. For comparison, let us examine some of the alternatives below:

- (i) Let $v(t) = u(-t)$ for $u(t) \in \mathcal{L}_2(-\infty, 0]$, and then $v(t) \in \mathcal{L}_2[0, \infty)$. Hence, the Hankel operator can be written as

$$\Gamma_g : \mathcal{L}_2[0, \infty) \mapsto \mathcal{L}_2[0, \infty) \quad \text{or} \quad \Gamma_G : \mathcal{H}_2 \mapsto \mathcal{H}_2$$

$$\begin{aligned} (\Gamma_g v)(t) &= \begin{cases} \int_0^\infty g(t+\tau)v(\tau)d\tau, & t \geq 0; \\ 0, & t < 0 \end{cases} \\ &= \int_0^\infty C e^{A(t+\tau)} B v(\tau) d\tau, \quad \text{for } t \geq 0. \end{aligned}$$

- (ii) In some applications, it is more convenient to work with an anticausal operator G and view the Hankel operator associated with G as the mapping from the future input to the past output. It will be clear in later chapters that this operator is closely related to the problem of approximating an anticausal function by a causal function, which is the problem at the heart of the $\mathcal{H}_\infty$ Control Theory.

Let $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array} \right]$ be an antistable transfer matrix, i.e., all the eigenvalues of A have positive real parts. Then the Hankel operator associated with $G(s)$ can be written as

$$\hat{\Gamma}_g : \mathcal{L}_2[0, \infty) \mapsto \mathcal{L}_2(-\infty, 0]$$

$$\begin{aligned} (\hat{\Gamma}_g v)(t) &= \begin{cases} \int_0^\infty g(t-\tau)v(\tau)d\tau, & t \leq 0; \\ 0, & t > 0 \end{cases} \\ &= \int_0^\infty C e^{A(t-\tau)} B v(\tau) d\tau, \quad \text{for } t \leq 0 \end{aligned}$$

or in the frequency domain

$$\hat{\Gamma}_G = P_- M_G|_{\mathcal{H}_2} : \mathcal{H}_2 \mapsto \mathcal{H}_2^\perp$$

$$\hat{\Gamma}_G v = P_-(Gv), \quad \text{for } v \in \mathcal{H}_2.$$

Now for any $v \in \mathcal{H}_2$ and $u \in \mathcal{H}_2^\perp$, we have

$$\langle P_-(Gv), u \rangle = \langle Gv, u \rangle = \langle v, G^\sim u \rangle = \langle v, P_+(G^\sim u) \rangle.$$

Hence, $\hat{\Gamma}_G = \Gamma_{G^\sim}^*$.

♡

8.2 All-pass Dilations

This section considers the dilation of a given transfer function to an all-pass transfer function. This transfer function dilation is the key to the optimal Hankel norm approximation in the next section. But first we need some preliminary results and some state space characterizations of all-pass functions.

Definition 8.1 *The inertia of a general complex, square matrix A denoted $\text{In}(A)$ is the triple $(\pi(A), \nu(A), \delta(A))$ where*

$\pi(A)$ = number of eigenvalues of A in the open right half-plane.

$\nu(A)$ = number of eigenvalues of A in the open left half-plane.

$\delta(A)$ = number of eigenvalues of A on the imaginary axis.

Theorem 8.2 *Given complex $n \times n$ and $n \times m$ matrices A and B , and hermitian matrix $P = P^*$ satisfying*

$$AP + PA^* + BB^* = 0 \quad (8.3)$$

then

$$(1) \text{ If } \delta(P) = 0 \text{ then } \pi(A) \leq \nu(P), \nu(A) \leq \pi(P).$$

$$(2) \text{ If } \delta(A) = 0 \text{ then } \pi(P) \leq \nu(A), \nu(P) \leq \pi(A).$$

Proof. (1) If $\delta(P) = 0$ then observe that (8.3) implies

$$AP + PA^* \leq 0.$$

Now suppose $\pi(A) > \nu(P)$ then there is an eigenvalue of A , $\lambda \in \mathbb{C}$ with $\operatorname{Re}(\lambda) > 0$, and a corresponding eigenvector, $x \in \mathbb{C}^n$, such that $x^*A = \lambda x^*$ and $x^*Px > 0$, which implies

$$x^*(AP + PA^*)x = (\lambda + \bar{\lambda})x^*Px \leq 0$$

i.e., $\operatorname{Re}(\lambda) \leq 0$, a contradiction. Hence $\pi(A) \leq \nu(P)$. Similarly, we can show that $\nu(A) \leq \pi(P)$.

(2) Assume $\delta(A) = 0$ and that $P = U \begin{bmatrix} P_1 & 0 \\ 0 & 0 \end{bmatrix} U^*$ with $\delta(P_1) = 0$, $U^*U = I$, and define

$$\tilde{A} = U^*AU = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad \tilde{B} = U^*B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}.$$

Then $U^*(8.3)U$ gives

$$\tilde{A} \begin{bmatrix} P_1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} P_1 & 0 \\ 0 & 0 \end{bmatrix} \tilde{A} + \tilde{B}\tilde{B}^* = 0 \quad (8.4)$$

$$(8.4) \Rightarrow B_2B_2^* = 0 \Rightarrow B_2 = 0 \quad (8.5)$$

$$(8.4), (8.5) \Rightarrow A_{21}P_1 = 0 \Rightarrow A_{21} = 0 \quad (8.6)$$

$$(8.4) \Rightarrow A_{11}P_1 + P_1A_{11}^* + B_1B_1^* = 0 \quad (8.7)$$

$$\Rightarrow (\text{by part (1)}) \quad \nu(A_{11}) \leq \pi(P_1)$$

$$\pi(A_{11}) \leq \nu(P_1)$$

but since $\delta(A_{11}) = \delta(P_1) = 0$

$$\pi(P_1) = \nu(A_{11}) \leq \nu(A)$$

$$\nu(P_1) = \pi(A_{11}) \leq \pi(A).$$

□

Theorem 8.3 *Given a realization (A, B, C) (not necessarily stable) with $A \in \mathbb{C}^{n \times n}$, $B \in \mathbb{C}^{n \times m}$, $C \in \mathbb{C}^{m \times n}$, then*

(1) If (A, B, C) is completely controllable and completely observable the following two statements are equivalent:

- (a) there exists a D such that $GG^\sim = \sigma^2 I$ where $G(s) = D + C(sI - A)^{-1}B$.
- (b) there exist $P, Q \in \mathbb{C}^{n \times n}$ such that
 - (i) $P = P^*, Q = Q^*$
 - (ii) $AP + PA^* + BB^* = 0$
 - (iii) $A^*Q + QA + C^*C = 0$
 - (iv) $PQ = \sigma^2 I$

(2) Given that part (1b) is satisfied then there exists a D satisfying

$$\begin{aligned} D^*D &= \sigma^2 I \\ D^*C + B^*Q &= 0 \\ DB^* + CP &= 0 \end{aligned}$$

and any such D will satisfy part (1a) (note, observability and controllability are not assumed).

Proof. Any systems satisfying part (1a) or (1b) can be transformed to the case $\sigma = 1$ by $\hat{B} = B/\sqrt{\sigma}$, $\hat{C} = C/\sqrt{\sigma}$, $\hat{D} = D/\sigma$, $\hat{P} = P/\sigma$, $\hat{Q} = Q/\sigma$. Hence, without loss of generality the proof will be given for the case $\sigma = 1$ only.

(1a) $\Rightarrow$ (1b) This is proved by constructing P and Q to satisfy (1b) as follows. Given (1a), $G(\infty) = D \Rightarrow DD^* = I$. Also $GG^\sim = I \Rightarrow G^\sim = G^{-1}$, i.e.,

$$\begin{aligned} G^{-1}(s) &= \left[\begin{array}{c|c} A - BD^{-1}C & -BD^{-1} \\ \hline D^{-1}C & D^{-1} \end{array} \right] = \left[\begin{array}{c|c} A - BD^*C & -BD^* \\ \hline D^*C & D^* \end{array} \right] \\ &= G^\sim = \left[\begin{array}{c|c} -A^* & -C^* \\ \hline B^* & D^* \end{array} \right]. \end{aligned}$$

These two transfer functions are identical and both minimal (since (A, B, C) is assumed to be minimal), and hence there exists a similarity transformation T relating the state-space descriptions, i.e.,

$$-A^* = T(A - BD^*C)T^{-1} \quad (8.8)$$

$$C^* = T B D^* \quad (8.9)$$

$$B^* = D^* C T^{-1}. \quad (8.10)$$

Further

$$(8.9) \Rightarrow B^* = D^* C (T^*)^{-1} \quad (8.11)$$

$$(8.10) \Rightarrow C^* = T^* B D^* \quad (8.12)$$

$$\begin{aligned} (8.8) \Rightarrow -A^* &= -C^* D B^* + (T^{-1} A^* T)^* \\ &= T^* (A - (T^*)^{-1} C^* D B^* T^*) (T^*)^{-1} \\ (8.9) \text{ and } (8.10) \Rightarrow &= T^* (A - B D^* C) (T^*)^{-1}. \end{aligned} \quad (8.13)$$

Hence, T and T^* satisfy identical equations, (8.8) to (8.10) and (8.11) to (8.13), and minimality implies these have a unique solution and hence $T = T^*$.

Now setting

$$Q = -T \quad (8.14)$$

$$P = -T^{-1} \quad (8.15)$$

clearly satisfies part (1b), equations (i) and (iv). Further, (8.8) and (8.9) imply

$$TA + A^*T - C^*C = 0 \quad (8.16)$$

which verifies (1b), equation (iii). Also (8.16) implies

$$AT^{-1} + T^{-1}A^* - T^{-1}C^*CT^{-1} = 0 \quad (8.17)$$

which together with (8.10) implies part (1b), equation (ii).

(1b) $\Rightarrow$ (1a) This is proved by first constructing D according to part (2) and then verifying part (1a) by calculation. Firstly note that since $Q = P^{-1}$, $Q \times ((1b), \text{equation (ii)}) \times Q$ gives

$$QA + A^*Q + QBB^*Q = 0 \quad (8.18)$$

which together with part (1b), equation (iii) implies that

$$QBB^*Q = C^*C \quad (8.19)$$

and hence by Lemma 2.14 there exists a D such that $D^*D = I$ and

$$DB^*Q = -C \quad (8.20)$$

$$DB^* = -CQ^{-1} = -CP. \quad (8.21)$$

Equations (8.20) and (8.21) imply that the conditions of part (2) are satisfied. Now note that

$$\begin{aligned} BB^* &= (sI - A)P + P(-sI - A^*) \\ \Rightarrow C(sI - A)^{-1}BB^*(-sI - A^*)^{-1}C^* &= CP(-sI - A^*)^{-1}C^* + C(sI - A)^{-1}PC^* \\ (8.21) \Rightarrow &= -DB^*(-sI - A^*)^{-1}C^* - C(sI - A)^{-1}BD^*. \end{aligned}$$

Hence, on expanding $G(s)G^\sim(s)$ we get

$$G(s)G^\sim = I.$$

Part (2) follows immediately from the proof of (1b) $\Rightarrow$ (1a) above. $\square$

The following theorem dilates a given transfer function to an all-pass function.

Theorem 8.4 Let $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]$ with $A \in \mathbb{C}^{n \times n}, B \in \mathbb{C}^{n \times m}, C \in \mathbb{C}^{m \times n}, D \in \mathbb{C}^{m \times m}$ satisfy

$$AP + PA^* + BB^* = 0 \quad (8.22)$$

$$A^*Q + QA + C^*C = 0 \quad (8.23)$$

for

$$P = P^* = \text{diag}(\Sigma_1, \sigma I_r) \quad (8.24)$$

$$Q = Q^* = \text{diag}(\Sigma_2, \sigma I_r) \quad (8.25)$$

with Σ_1 and Σ_2 diagonal, $\sigma \neq 0$ and $\delta(\Sigma_1 \Sigma_2 - \sigma^2 I) = 0$.

Partition (A, B, C) conformably with P , as

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad C = \begin{bmatrix} C_1 & C_2 \end{bmatrix} \quad (8.26)$$

and define $W(s) := \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array} \right]$ with

$$\hat{A} = \Gamma^{-1}(\sigma^2 A_{11}^* + \Sigma_2 A_{11} \Sigma_1 - \sigma C_1^* U B_1^*) \quad (8.27)$$

$$\hat{B} = \Gamma^{-1}(\Sigma_2 B_1 + \sigma C_1^* U) \quad (8.28)$$

$$\hat{C} = C_1 \Sigma_1 + \sigma U B_1^* \quad (8.29)$$

$$\hat{D} = D - \sigma U \quad (8.30)$$

where U is a unitary matrix satisfying

$$B_2 = -C_2^* U \quad (8.31)$$

and

$$\Gamma = \Sigma_1 \Sigma_2 - \sigma^2 I. \quad (8.32)$$

Also define the error system

$$E(s) = G(s) - W(s) = \left[\begin{array}{c|c} A_e & B_e \\ \hline C_e & D_e \end{array} \right]$$

with

$$A_e = \begin{bmatrix} A & 0 \\ 0 & \hat{A} \end{bmatrix}, \quad B_e = \begin{bmatrix} B \\ \hat{B} \end{bmatrix}, \quad C_e = \begin{bmatrix} C & -\hat{C} \end{bmatrix}, \quad D_e = D - \hat{D}. \quad (8.33)$$

Then

(1) (A_e, B_e, C_e) satisfy

$$A_e P_e + P_e A_e^* + B_e B_e^* = 0 \quad (8.34)$$

$$A_e^* Q_e + Q_e A_e + C_e^* C_e = 0 \quad (8.35)$$

with

$$P_e = \begin{bmatrix} \Sigma_1 & 0 & I \\ 0 & \sigma I_r & 0 \\ I & 0 & \Sigma_2 \Gamma^{-1} \end{bmatrix} \quad (8.36)$$

$$Q_e = \begin{bmatrix} \Sigma_2 & 0 & -\Gamma \\ 0 & \sigma I_r & 0 \\ -\Gamma & 0 & \Sigma_1 \Gamma \end{bmatrix} \quad (8.37)$$

$$P_e Q_e = \sigma^2 I \quad (8.38)$$

(2) $E(s)E^\sim(s) = \sigma^2 I$.

(3) If $\delta(A) = 0$ then

(a) $\delta(\hat{A}) = 0$

(b) If $\delta(\Sigma_1 \Sigma_2) = 0$ then

$$\text{In}(\hat{A}) = \text{In}(-\Sigma_1 \Gamma) = \text{In}(-\Sigma_2 \Gamma)$$

(c) If $P > 0, Q > 0$ then the McMillan degree of the stable part of $(\hat{A}, \hat{B}, \hat{C})$ equals $\pi(\Sigma_1 \Gamma) = \pi(\Sigma_2 \Gamma)$.

(d) If either (i) $\Sigma_1 \Gamma > 0$ and $\Sigma_2 \Gamma > 0$ or (ii) $\Sigma_1 \Gamma < 0$ and $\Sigma_2 \Gamma < 0$ then $(\hat{A}, \hat{B}, \hat{C})$ is a minimal realization.

Proof. For notational convenience it will be assumed that $\sigma = 1$ and this can be done without loss of generality since B, C and Σ can be simply rescaled to give $\sigma = 1$.

It is first necessary to verify that there exists a unitary matrix U satisfying (8.31). The (2,2) blocks of (8.22) and (8.23) give

$$A_{22} + A_{22}^* + B_2 B_2^* = 0 \quad (8.39)$$

$$A_{22}^* + A_{22} + C_2^* C_2 = 0 \quad (8.40)$$

and hence $B_2 B_2^* = C_2^* C_2$ and by Lemma 2.14 there exists a unitary U satisfying (8.31).

(1) The proof of equations (8.34) to (8.38) is by a straightforward calculation, as follows. To verify (8.34) and (8.36) we need (8.22) which is assumed, together with

$$A_{11} + \hat{A}^* + B_1 \hat{B}^* = 0 \quad (8.41)$$

$$A_{21} + B_2 \hat{B}^* = 0 \quad (8.42)$$

$$\hat{A} \Sigma_2 \Gamma^{-1} + \Sigma_2 \Gamma^{-1} \hat{A}^* + \hat{B} \hat{B}^* = 0 \quad (8.43)$$

which will now be verified.

$$B_2 \hat{B}^* = B_2(B_1^* \Sigma_2 + U^* C_1) \Gamma^{-1} \quad (8.44)$$

$$(8.31) \Rightarrow = (B_2 B_1^* \Sigma_2 - C_2^* C_1) \Gamma^{-1} \quad (8.45)$$

$$\begin{aligned} &= ((-A_{21} \Sigma_1 - A_{12}^*) \Sigma_2 + A_{12}^* \Sigma_2 + A_{21}) \Gamma^{-1} \quad (8.46) \\ &= -A_{21} \Rightarrow (8.42) \end{aligned}$$

where $B_2 B_1^*$ and $C_2^* C_1$ were substituted in (8.45) using the (2,1) blocks of (8.22) and (8.23), respectively. To verify (8.41)

$$B_1 \hat{B}^* = (B_1 B_1^* \Sigma_2 + B_1 U^* C_1) \Gamma^{-1} \quad (8.47)$$

$$\begin{aligned} &= (-A_{11} \Sigma_1 \Sigma_2 - \Sigma_1 A_{11}^* \Sigma_2 + B_1 U^* C_1) \Gamma^{-1} \quad (8.48) \\ &= -A_{11} - \hat{A}^* \Rightarrow (8.41) \end{aligned}$$

where $B_1 B_1^*$ was substituted using the (1,1) block of (8.22) and (8.27) substituting in (8.48). Finally to verify (8.43) consider

$$\Gamma \hat{A} \Sigma_2 + \Sigma_2 \hat{A}^* \Gamma = (A_{11}^* + \Sigma_2 A_{11} \Sigma_1 - C_1^* U B_1^*) \Sigma_2 + \Sigma_2 (A_{11} + \Sigma_1 A_{11}^* \Sigma_2 - B_1 U^* C_1) \quad (8.49)$$

$$\begin{aligned} &= -(\Sigma_2 B_1 + C_1^* U)(B_1^* \Sigma_2 + U^* C_1) + (A_{11}^* \Sigma_2 + \Sigma_2 A_{11} + C_1^* C_1) \\ &\quad + \Sigma_2 (A_{11} \Sigma_1 + \Sigma_1 A_{11}^* + B_1 B_1^*) \Sigma_2 \quad (8.50) \end{aligned}$$

$$= -\Gamma \hat{B} \hat{B}^* \Gamma \Rightarrow (8.43)$$

where (8.27) $\rightarrow$ (8.49) and (8.50) is a rearrangement of (8.49) and finally, the (1,1) blocks of (8.22) and (8.23) are used. Equations (8.34) and (8.36) are hence verified.

Similarly in order to verify (8.35) and (8.37) we need (8.23) which is assumed together with

$$A_{11}^* (-\Gamma) + (-\Gamma) \hat{A} - C_1^* \hat{C} = 0 \quad (8.51)$$

$$A_{12}^* (-\Gamma) - C_2^* \hat{C} = 0 \quad (8.52)$$

$$\hat{A}^* \Sigma_1 \Gamma + \Sigma_1 \Gamma \hat{A} + \hat{C}^* \hat{C} = 0. \quad (8.53)$$

Equations (8.51) to (8.53) are now verified in an analogous manner to equations (8.41) to (8.43)

$$\begin{aligned} C_2^* \hat{C} &= C_2^* (C_1 \Sigma_1 + U B_1^*) \\ &= (-A_{12}^* \Sigma_2 - A_{21}) \Sigma_1 - B_2 B_1^* \\ &= -A_{12}^* \Gamma \Rightarrow (8.52) \\ C_1^* \hat{C} &= C_1^* C_1 \Sigma_1 + C_1^* U B_1^* \\ &= -A_{11}^* \Sigma_2 \Sigma_1 - \Sigma_2 A_{11} \Sigma_1 + C_1^* U B_1^* \end{aligned}$$

$$\begin{aligned}
&= -A_{11}^* \Gamma - \Gamma \hat{A} \Rightarrow (8.51) \\
\hat{A}^* \Sigma_1 \Gamma + \Sigma_1 \Gamma \hat{A} &= (A_{11} + \Sigma_1 A_{11}^* \Sigma_2 - B_1 U^* C_1) \Sigma_1 + \Sigma_1 (A_{11}^* + \Sigma_2 A_{11} \Sigma_1 - C_1^* U B_1) \\
&= -(\Sigma_1 C_1^* + B_1 U^*) (C_1 \Sigma_1 + U B_1^*) \\
&\quad + \Sigma_1 (A_{11}^* \Sigma_2 + \Sigma_2 A_{11} + C_1^* C_1) \Sigma_1 + (A_{11} \Sigma_1 + \Sigma_1 A_{11}^* + B_1 B_1^*) \\
&= -\hat{C}^* \hat{C} \Rightarrow (8.53)
\end{aligned}$$

Therefore, (8.35) and (8.37) have been verified, (8.38) is immediate, and the proof of part (1) is complete.

(2) Equations (8.34), (8.35) and (8.38) ensure the conditions of Theorem 8.3, part (1b) are satisfied and Theorem 8.3, part (2) can be used to show that the D_e given in (8.33) makes $E(s)$ all-pass. (Note it is still assumed that $\sigma = 1$.) We hence need to verify that

$$D_e^* D_e = I \quad (8.54)$$

$$D_e^* C_e + B_e^* Q_e = 0 \quad (8.55)$$

$$D_e B_e^* + C_e P_e = 0. \quad (8.56)$$

Equation (8.54) is immediate, (8.55) follows by substituting the definitions of $\hat{B}, \hat{C}, D_e$ and Q , and (8.56) follows from $D_e \times (8.55) \times P_e$.

(3) (a) To show that $\delta(\hat{A}) = 0$ if $\delta(A) = 0$ we will assume that there exists $x \in \mathbb{C}^{n-r}$ and $\lambda \in \mathbb{C}$ such that $\hat{A}x = \lambda x$ and $\lambda + \bar{\lambda} = 0$, and show that this implies $x = 0$. From $x^*(8.53)x$,

$$(\lambda + \bar{\lambda})x^* \Sigma_1 \Gamma x + x^* \hat{C}^* \hat{C} x = 0 \quad (8.57)$$

$$\Rightarrow \hat{C}x = 0. \quad (8.58)$$

Now (8.51) x gives

$$\begin{aligned}
-A_{11}^* \Gamma x - \Gamma \lambda x + C_1^* \hat{C}x &= 0 \\
\Rightarrow x^* \Gamma A_{11} &= -\bar{\lambda} x^* \Gamma.
\end{aligned} \quad (8.59)$$

Also (8.52) x and (8.58) give

$$A_{12}^* \Gamma x = 0. \quad (8.60)$$

Equations (8.59) and (8.60) imply that $(x^* \Gamma, 0)A = -\bar{\lambda}(x^* \Gamma, 0)$ but since it is assumed that $\delta(A) = 0$, $\lambda + \bar{\lambda} = 0$ and Γ^{-1} exists this implies that $x = 0$ and $\delta(\hat{A}) = 0$ is proven.

(b) Since $\delta(\hat{A}) = 0$ has been proved and $\delta(\Sigma_1 \Sigma_2) = 0$ is assumed ($\Rightarrow \delta(\Sigma_2 \Gamma^{-1}) = \delta(\Sigma_1 \Gamma) = 0$) Theorem 8.2 can be applied since equations (8.43) and (8.53) have been verified. Hence

$$\text{In}(\hat{A}) = \text{In}(-\Sigma_1 \Gamma^{-1}) = \text{In}(-\Sigma_1 \Gamma) = \text{In}(-\Sigma_2 \Gamma)$$

(c) Assume that there exists $x \neq 0 \in \mathbb{C}^{n-r}$ and $\lambda \in \mathbb{C}$ such that $\hat{A}x = \lambda x$ and $\hat{C}x = 0$ (i.e., $(\hat{C}, \hat{A})$ is not completely observable). Then (8.51) x and (8.52) x give

$$\begin{aligned}
-A_{11}^* \Gamma x - \Gamma \lambda x &= 0 \\
-A_{12}^* \Gamma x &= 0
\end{aligned}$$

hence $(-\bar{\lambda})$ is an eigenvalue of A since $\Gamma x \neq 0$. However, since $P > 0$ and $\delta(A) = 0$ are assumed then $\text{In}(A) = (0, n, 0)$ and all the unobservable modes must be in the open right half plane. Similarly, if it is assumed that $(\hat{A}, \hat{B})$ is not completely controllable then (8.41) and (8.42) will give the analogous conclusion and therefore all the modes in the left half-plane are controllable and observable, and the condition in (3b) gives their number.

(d) (i) If $\Sigma_1 \Gamma > 0$ or $\Sigma_2 \Gamma > 0$ then by (3b) $\text{In}(\hat{A}) = (0, n - r, 0)$ and by (3c) the McMillan degree of $(\hat{A}, \hat{B}, \hat{C})$ is $n - r$ and the result is proven.

(ii) Assume there exists x such that $\hat{A}x = \lambda x$ and $\hat{C}x = 0$. Then $x^*(8.53)x$ gives

$$(\lambda + \bar{\lambda})x^* \Sigma_1 \Gamma x = 0$$

but $(\lambda + \bar{\lambda}) \neq 0$ by (3a) and $\Sigma_1 \Gamma < 0$ is assumed so that $x = 0$. Hence $(\hat{C}, \hat{A})$ is completely observable. Similarly (8.43) gives $(\hat{A}, \hat{B})$ completely controllable. $\square$

Example 8.1 Take

$$G(s) = \frac{39s^2 + 105s + 250}{(s+2)(s+5)^2}.$$

This has a balanced realization given by

$$A = \begin{bmatrix} -2 & 4 & -4 \\ -4 & -1 & -4 \\ -4 & 4 & -9 \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 1 \\ 6 \end{bmatrix}, \quad C = [2 \quad -1 \quad 6], \quad \Sigma = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Now using the above construction with $\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$, $\sigma = 2$ gives

$$\hat{A} = \frac{1}{3} \begin{bmatrix} 2 & 10 \\ -8 & 5 \end{bmatrix}, \quad \hat{B} = \frac{1}{3} \begin{bmatrix} 2 \\ -2 \end{bmatrix}, \quad \hat{C} = [-2 \quad -5/2], \quad \hat{D} = 2$$

$$W(s) = \frac{6s^2 - 13s + 90}{3s^2 - 7s + 30}, \quad G(s) - W(s) = \frac{2(-s+2)(-s+5)^2(3s^2+7s+30)}{(s+2)(s+5)^2(3s^2-7s+30)}$$

$W(s)$ is an optimal anticausal approximation to $G(s)$ with $\mathcal{L}_\infty$ error of 2. $\diamond$

Example 8.2 Let us also illustrate Theorem 8.4 when $(\Sigma_1^2 - \sigma^2 I)$ is indefinite. Take $G(s)$ as in the above example and permute the first and third states of the balanced realization so that $\Sigma = \text{diag}(2, \frac{1}{2}, 1)$, $\Sigma_1 = \text{diag}(2, \frac{1}{2})$, $\sigma = 1$. The construction of Theorem 8.4 now gives

$$\hat{A} = \begin{bmatrix} -3 & 2 \\ 8 & 3 \end{bmatrix}, \quad \hat{B} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}, \quad \hat{C} = [6 \quad -3/2], \quad \hat{D} = 1.$$

Theorem 8.4, part (2b) implies that

$$\text{In}(\hat{A}) = \text{In}(-\Sigma_1(\Sigma_1^2 - \sigma^2 I)) = \text{In} \begin{bmatrix} -6 & 0 \\ 0 & \frac{3}{8} \end{bmatrix} = (1, 1, 0)$$

which is verified by noting that $\hat{A}$ has eigenvalues of 5 and -5 .

$$W(s) = \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array} \right] = \frac{s+20}{s+5}$$

and we note that the stable part of $W(s)$ has McMillan degree 1 as predicted by Theorem 8.4, part (3c). However, this example has been constructed to show that $(\hat{A}, \hat{B}, \hat{C})$ itself may not be minimal when the conditions of part (3d) are not satisfied, and in this case the unstable pole at $+5$ is both uncontrollable and unobservable. $\frac{s+20}{s+5}$ is in fact an optimal Hankel norm approximation to $G(s)$ of degree 1 and

$$E(s) = \frac{(-s+2)(-s+5)^2}{(s+2)(s+5)^2}.$$

In general the error $E(j\omega)$ will have modulus equal to σ but $E(s)$ will contain unstable poles. $\diamond$

Example 8.3 Let us finally complete the analysis of this $G(s)$ by permuting the second and third states in the balanced realization of the last example to obtain $\Sigma_1 = \text{diag}(2, 1)$, $\sigma = \frac{1}{2}$. We will find

$$\hat{A} = \begin{bmatrix} -15 & -4 \\ -20 & -6 \end{bmatrix}, \quad \hat{B} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}, \quad \hat{C} = \begin{bmatrix} 15 & 3 \end{bmatrix}, \quad \hat{D} = -\frac{1}{2}$$

$$W(s) = \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array} \right] = \frac{1}{2} \frac{(-s^2 + 123s + 110)}{(s^2 + 21s + 10)}$$

$$E(s) = G(s) - W(s) = -\frac{1}{2} \frac{(-s+2)(-s+5)^2(s^2 - 21s + 10)}{(s+2)(s+5)^2(s^2 + 21s + 10)}.$$

Note that $-\Sigma_1\Gamma = \text{diag}(-15/2, -3/4)$ so that $\hat{A}$ is stable by Theorem 8.4, part (3b). $|E(j\omega)| = \frac{1}{2}$ by Theorem 8.4, part (2), $(\hat{A}, \hat{B}, \hat{C})$ is minimal by Theorem 8.4, part (3d). $W(s)$ is in fact an optimal second-order Hankel norm approximation to $G(s)$. $\diamond$

8.3 Optimal Hankel Norm Approximation

We are now ready to give a solution to the optimal Hankel norm approximation problem based on Theorem 8.4. The following Lemma gives a lower bound on the achievable Hankel norm of the error

$$\inf_{\hat{G}} \left\| G(s) - \hat{G}(s) \right\|_H \geq \sigma_{k+1}(G)$$

and then Theorem 8.6 shows that the construction of Theorem 8.4 can be used to achieve this lower bound.

Lemma 8.5 *Given a stable, rational, $p \times m$, transfer function matrix $G(s)$ with Hankel singular values $\sigma_1 \geq \sigma_2 \dots \geq \sigma_k \geq \sigma_{k+1} \geq \sigma_{k+2} \dots \geq \sigma_n > 0$, then for all $\hat{G}(s)$ stable and of McMillan degree $\leq k$*

$$\sigma_i(G(s) - \hat{G}(s)) \geq \sigma_{i+k}(G(s)), \quad i = 1, \dots, n-k, \quad (8.61)$$

$$\sigma_{i+k}(G(s) - \hat{G}(s)) \leq \sigma_i(G(s)), \quad i = 1, \dots, n. \quad (8.62)$$

In particular,

$$\|G(s) - \hat{G}(s)\|_H \geq \sigma_{k+1}(G(s)). \quad (8.63)$$

Proof. We shall prove (8.61) only and the inequality (8.62) follows from (8.61) by setting

$$G(s) = (G(s) - \hat{G}(s)) - (-\hat{G}(s)).$$

Let $(\hat{A}, \hat{B}, \hat{C})$ be a minimal state space realization of $\hat{G}(s)$, then (A_e, B_e, C_e) given by (8.33) will be a state space realization of $G(s) - \hat{G}(s)$. Now let $P = P^*$ and $Q = Q^*$ satisfy (8.34) and (8.35) respectively (but not necessary (8.36) and (8.37) and write

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{12}^* & P_{22} \end{bmatrix}, \quad Q = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{12}^* & Q_{22} \end{bmatrix}, \quad P_{11}, Q_{11} \in \mathbb{R}^{n \times n}.$$

Since $P \geq 0$ it can be factorized as

$$P = RR^*$$

where

$$R = \begin{bmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{bmatrix}$$

with

$$R_{22} = P_{22}^{1/2}, \quad R_{12} = P_{12}P_{22}^{-1/2}, \quad R_{11}R_{11}^* = P_{11} - R_{12}R_{12}^*$$

($P_{22} > 0$ since $(\hat{A}, \hat{B}, \hat{C})$ is a minimal realization.)

$$\begin{aligned} \sigma_i(G(s) - \hat{G}(s)) &= \lambda_i(PQ) = \lambda_i(RR^*Q) = \lambda_i(R^*QR) \\ &\geq \lambda_i \left(\begin{bmatrix} I_n & 0 \end{bmatrix} R^*QR \begin{bmatrix} I_n \\ 0 \end{bmatrix} \right) \\ &= \lambda_i \left(\begin{bmatrix} R_{11}^* & 0 \end{bmatrix} Q \begin{bmatrix} R_{11} \\ 0 \end{bmatrix} \right) \\ &= \lambda_i(R_{11}^*Q_{11}R_{11}) = \lambda_i(Q_{11}R_{11}R_{11}^*) \\ &= \lambda_i(Q_{11}(P_{11} - R_{12}R_{12}^*)) \\ &= \lambda_i(Q_{11}^{1/2}P_{11}Q_{11}^{1/2} - XX^*) \quad \text{where } X = Q_{11}^{1/2}R_{12} \\ &\geq \lambda_{i+k}(Q_{11}^{1/2}P_{11}Q_{11}^{1/2}) \\ &= \lambda_{i+k}(P_{11}Q_{11}) = \sigma_{i+k}^2(G) \end{aligned} \quad (8.64)$$

where (8.64) follows from the fact that X is an $n \times k$ matrix ($\Rightarrow \text{rank}(XX^*) \leq k$). $\square$

We can now give a solution to the optimal Hankel norm approximation problem for square transfer functions.

Theorem 8.6 *Given a stable, rational, $m \times m$, transfer function $G(s)$ then*

$$(1) \quad \sigma_{k+1}(G(s)) = \inf_{\hat{G} \in \mathcal{H}_\infty, F \in \mathcal{H}_\infty^-} \|G(s) - \hat{G}(s) - F(s)\|_\infty, \text{ McMillan degree } (\hat{G}) \leq k.$$

(2) *If $G(s)$ has Hankel singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k > \sigma_{k+1} = \sigma_{k+2} = \dots = \sigma_{k+r} > \sigma_{k+r+1} \geq \dots \geq \sigma_n > 0$ then $\hat{G}(s)$ of McMillan degree k is an optimal Hankel norm approximation to $G(s)$ if and only if there exists $F(s) \in \mathcal{H}_\infty^-$ (whose McMillan degree can be chosen $\leq n + k - 1$) such that $E(s) := G(s) - \hat{G}(s) - F(s)$ satisfies*

$$E(s)E^\sim(s) = \sigma_{k+1}^2 I. \quad (8.65)$$

In which case

$$\|G(s) - \hat{G}(s)\|_H = \sigma_{k+1}. \quad (8.66)$$

(3) *Let $G(s)$ be as in (2) above, then an optimal Hankel norm approximation of McMillan degree k , $\hat{G}(s)$, can be constructed as follows. Let (A, B, C) be a balanced realization of $G(s)$ with corresponding*

$$\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_k, \sigma_{k+r+1}, \dots, \sigma_n, \sigma_{k+1}, \dots, \sigma_{k+r}),$$

and define $(\hat{A}, \hat{B}, \hat{C}, \hat{D})$ from equations (8.26) to (8.31). Then

$$\hat{G}(s) + F(s) = \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array} \right] \quad (8.67)$$

where $\hat{G}(s) \in \mathcal{H}_\infty$ and $F(s) \in \mathcal{H}_\infty^-$ with the McMillan degree of $\hat{G}(s) = k$ and the McMillan degree of $F(s) = n - k - r$.

Proof. By the definition of $\mathcal{L}_\infty$ norm, for all $F(s) \in \mathcal{H}_\infty^-$, and $\hat{G}(s)$ of McMillan degree k

$$\begin{aligned} \|G(s) - \hat{G}(s) - F(s)\|_\infty &\geq \sup_{f \in \mathcal{H}_2^\perp, \|f\|_2 \leq 1} \|(G(s) - \hat{G}(s) - F(s))f\|_2 \\ &\geq \sup_{f \in \mathcal{H}_2^\perp, \|f\|_2 \leq 1} \|P_+(G - \hat{G} - F)f\|_2 \\ &= \sup_{f \in \mathcal{H}_2^\perp, \|f\|_2 \leq 1} \|P_+(G - \hat{G})f\|_2 \\ &= \|G - \hat{G}\|_H \\ &\geq \sigma_{k+1}(G(s)) \end{aligned} \quad (8.68)$$

where (8.68) follows from Lemma 8.5.

Now define $\hat{G}(s)$ and $F(s)$ via equation (8.26), then Theorem 8.4, part (2) implies that (8.65) holds and hence

$$\|E(s)\|_{\infty} = \sigma_{k+1}. \quad (8.69)$$

Also from Theorem 8.4, part (3b)

$$\text{In}(\hat{A}) = \text{In}(-\Sigma_1(\Sigma_1^2 - \sigma_{k+1}^2 I)) = (n - k - r, k, 0). \quad (8.70)$$

Hence, $\hat{G}$ has McMillan degree k and it is in the correct class, and therefore (8.69) implies that the inequalities in (8.68) become equalities, and part (1) is proven, as in part (3). Clearly the sufficiency of part (2) can be similarly verified by noting that (8.65) implies that (8.68) is satisfied with equality.

To show the necessity of part (2) suppose that $\hat{G}(s)$ is an optimal Hankel norm approximation to $G(s)$ of McMillan degree k , i.e., equation (8.66) holds. Now Theorem 8.4 can be applied to $G(s) - \hat{G}(s)$ to produce an optimal anticausal approximation $F(s)$, such that $(G(s) - \hat{G}(s) - F(s))/\sigma_{k+1}(G)$ is all-pass since $\sigma_{k+1}(G) = \sigma_1(G - \hat{G})$. Further, the McMillan degree of this $F(s)$ will be, the McMillan degree of $(G(s) - \hat{G}(s))$ minus the multiplicity of $\sigma_1(G - \hat{G})$, $\leq n + k - 1$. $\square$

The following corollary gives the solution to the well-known Nehari's problem.

Corollary 8.7 *Let $G(s)$ be a stable, rational, $m \times m$, transfer function of McMillan degree n such that $\sigma_1(G)$ has multiplicity r_1 . Also let $F(s)$ be an optimal anticausal approximation of degree $n - r_1$ given by the construction of Theorem 8.4. Then*

- (1) $(G(s) - F(s))/\sigma_1$ is all-pass.
- (2) $\sigma_{i-r_1}(F(-s)) = \sigma_i(G(s)), i = r_1 + 1, \dots, n$.

Proof. (1) is proved in Theorem 8.4, part (2). (2) is obtained from the forms of P_e and Q_e in Theorem 8.4, part (1). $F(-s)$ is used since it will be stable and have well-defined Hankel singular values. $\square$

The optimal Hankel norm approximation for non-square case can be obtained by first augmenting the function to form a square function. For example, consider a stable, rational, $p \times m$ ($p < m$), transfer function $G(s)$. Let $G_a = \begin{bmatrix} G \\ 0 \end{bmatrix}$ be an augmented square transfer function and let $\hat{G}_a = \begin{bmatrix} \hat{G} \\ \hat{G}_2 \end{bmatrix}$ be the optimal Hankel norm approximation of G_a such that

$$\|G_a - \hat{G}_a\|_H = \sigma_{k+1}(G_a).$$

Then

$$\sigma_{k+1}(G) \leq \|G - \hat{G}\|_H \leq \|G_a - \hat{G}_a\|_H = \sigma_{k+1}(G_a) = \sigma_{k+1}(G)$$

i.e., $\hat{G}$ is an optimal Hankel norm approximation of $G(s)$.

8.4 $\mathcal{L}_\infty$ Bounds for Hankel Norm Approximation

The natural question that arise now is, does the Hankel norm being small imply that any other more familiar norms are also small? We shall have a definite answer in this section.

Lemma 8.8 *Let an $m \times m$ transfer matrix $E = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]$ satisfy $E(s)E^\sim(s) = \sigma^2 I$ and all equations of Theorem 8.3 and let A have dimension $n_1 + n_2$ with n_1 eigenvalues strictly in the left half plane and $n_2 < n_1$ eigenvalues strictly in the right half plane. If $E = G_s + F$ with $G_s \in \mathcal{RH}_\infty$ and $F \in \mathcal{RH}_\infty^-$ then,*

$$\sigma_i(G_s) = \begin{cases} \sigma & i = 1, 2, \dots, n_1 - n_2 \\ \sigma_{i-n_1+n_2}(F(-s)) & i = n_1 - n_2 + 1, \dots, n_1 \end{cases}$$

Proof. Firstly let the realization be transformed to,

$$E = \left[\begin{array}{cc|c} A_1 & 0 & B_1 \\ 0 & A_2 & B_2 \\ \hline C_1 & C_2 & D \end{array} \right] = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right], \quad \operatorname{Re} \lambda_i(A_1) < 0, \quad \operatorname{Re} \lambda_i(A_2) > 0,$$

in which case $G = \left[\begin{array}{c|c} A_1 & B_1 \\ \hline C_1 & D \end{array} \right]$, $F = \left[\begin{array}{c|c} A_2 & B_2 \\ \hline C_2 & 0 \end{array} \right]$. The equations of Theorem 8.3 (i)-(iv) are then satisfied by a transformed P and Q , partitioned as,

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{12}^* & P_{22} \end{bmatrix}, \quad Q = \begin{bmatrix} Q_{11} & Q_{21}^* \\ Q_{21} & Q_{22} \end{bmatrix}$$

$PQ = \sigma^2 I$ implies that,

$$\begin{aligned} \det(\lambda I - P_{11}Q_{11}) &= \det(\lambda I - (\sigma^2 I - P_{12}Q_{21})) \\ &= \det((\lambda - \sigma^2)I + P_{12}Q_{21}) \\ &= (\lambda - \sigma^2)^{n_1-n_2} \det((\lambda - \sigma^2)I + Q_{21}P_{12}) \\ &= (\lambda - \sigma^2)^{n_1-n_2} \det(\lambda I - Q_{22}P_{22}). \end{aligned}$$

The result now follows on observing that $\sigma_i(G(s)) = \lambda_i(P_{11}Q_{11})$ and $\sigma_i^2(F(-s)) = \lambda_i(Q_{22}P_{22})$. $\square$

Corollary 8.9 *Let $E(s) = G(s) - \hat{G}(s) - F(s)$ be as defined in part (3) of Theorem 8.6 with $G(s), \hat{G}(s) \in \mathcal{RH}_\infty$ and $F(s) \in \mathcal{RH}_\infty^-$. Then for $i = 1, 2, \dots, 2k + r$,*

$$\sigma_i(G - \hat{G}) = \sigma_{k+1}(G),$$

and for $i = 1, 2, \dots, n - k - r$,

$$\sigma_{i+3k+r}(G) \leq \sigma_i(F(-s)) = \sigma_{i+2k+r}(G - \hat{G}) \leq \sigma_{i+k+r}(G)$$

Proof. The construction of $E(s)$ ensures that the all-pass equations are satisfied and an inertia argument easily establishes that the A -matrix has precisely $n_1 = n + k$ eigenvalues in the open left half plane and $n_2 = n - k - r$ in the open right half plane. Hence Lemma 8.8 can be applied to give the equalities. The inequalities follow from Lemma 8.5. $\square$

The following lemma gives the properties of certain optimal Hankel norm approximations when the degree is reduced by the multiplicity of σ_n . In this case some precise statements on the error and the approximating system can be made.

Lemma 8.10 *Let $G(s)$ be a stable, rational $m \times m$, transfer function of McMillan degree n and such that $\sigma_n(G)$ has multiplicity r . Also let $\hat{G}(s)$ be an optimal Hankel norm approximation of degree $n - r$ given by Theorem 8.6, part (3) (with $F(s) \equiv 0$) then*

$$(1) \quad (G(s) - \hat{G}(s))/\sigma_n(G(s)) \text{ is all-pass.}$$

$$(2) \quad \sigma_i(\hat{G}(s)) = \sigma_i(G(s)), i = 1, \dots, n - r.$$

Proof. Theorem 8.6 gives that $\hat{A} \in \mathbb{R}^{(n-r) \times (n-r)}$ is stable and hence $F(s)$ can be chosen to be zero and therefore $(G(s) - \hat{G}(s))/\sigma_n(G)$ is all-pass. The $\sigma_i(\hat{G}(s))$ are obtained from Lemma 8.8. $\square$

Applying the above reduction procedure again on $\hat{G}(s)$ and repeating until $\hat{G}(s)$ has zero McMillan degree gives the following new representation of stable systems.

Theorem 8.11 *Let $G(s)$ be a stable, rational $m \times m$, transfer function with Hankel singular values $\sigma_1 > \sigma_2 > \dots > \sigma_N$ where σ_i has multiplicity r_i and $r_1 + r_2 + \dots + r_N = n$. Then there exists a representation of $G(s)$ as*

$$G(s) = D_0 + \sigma_1 E_1(s) + \sigma_2 E_2(s) + \dots + \sigma_N E_N(s) \quad (8.71)$$

where

$$(1) \quad E_k(s) \text{ are all-pass and stable for all } k.$$

$$(2) \quad \text{For } k = 1, 2, \dots, N$$

$$\hat{G}_k(s) := D_0 + \sum_{i=1}^k \sigma_i E_i(s)$$

has McMillan degree $r_1 + r_2 + \dots + r_k$.

Proof. Let $\hat{G}_k(s)$ be the optimal Hankel norm approximation to $\hat{G}_{k+1}(s)$ (given by Lemma 8.10) of degree $r_1 + r_2 + \dots + r_k$, with $\hat{G}_N(s) := G(s)$. Lemma 8.10 (2) applied at each step then gives that the Hankel singular values of $\hat{G}_k(s)$ will be $\sigma_1, \sigma_2, \dots, \sigma_k$ with multiplicities $r_1, r_2, \dots, r_k$, respectively. Hence Lemma 8.10 (1) gives that $\hat{G}_k(s) - \hat{G}_{k-1}(s) = \sigma_k E_k(s)$ for some stable, all-pass $E_k(s)$. Note also that Theorem 8.4, part (3d), relation (i) also ensures that each $\hat{G}_k(s)$ will have McMillan degree $r_1 + r_2 + \dots + r_k$. Finally taking $D_0 = \hat{G}_0(s)$ which will be a constant and combining the steps gives the result. $\square$

Note that the construction of Theorem 8.11 immediately gives an approximation algorithm that will satisfy $\|G(s) - \hat{G}(s)\|_\infty \leq \sigma_{k+1} + \sigma_{k+2} + \dots + \sigma_N$. This will not be an optimal Hankel norm approximation in general, but would involve less computation since the decomposition into $\hat{G}(s) = F(s)$ need not be done, and at each step a balanced realization of $\hat{G}_k(s)$ is given by $(\hat{A}_k, \hat{B}_k, \hat{C}_k)$ with a diagonal scaling.

An upper-bound on the $\mathcal{L}_\infty$ norm of $G(s)$ is now obtained as an immediate consequence of Theorem 8.11.

Corollary 8.12 *Let $G(s)$ be a stable, rational $p \times m$, transfer function with Hankel singular values $\sigma_1 > \sigma_2 > \dots > \sigma_N$, where each σ_i has multiplicity r_i , and such that $G(\infty) = 0$. Then*

$$(1) \|G(s)\|_\infty \leq 2(\sigma_1 + \sigma_2 + \dots + \sigma_N)$$

$$(2) \text{ there exists a constant } D_0 \text{ such that}$$

$$\|G(s) - D_0\|_\infty \leq \sigma_1 + \sigma_2 + \dots + \sigma_N.$$

Proof. For $p = m$ consider the representation of $G(s)$ given by Theorem 8.11 then

$$\begin{aligned} \|G(s) - D_0\|_\infty &= \|\sigma_1 E_1(s) + \sigma_2 E_2(s) + \dots + \sigma_N E_N(s)\|_\infty \\ &\leq \sigma_1 + \sigma_2 + \dots + \sigma_N \end{aligned}$$

since $E_k(s)$ are all-pass. Further setting $s = \infty$, since $G(\infty) = 0$, gives

$$\begin{aligned} \|D_0\| &\leq \sigma_1 + \sigma_2 + \dots + \sigma_N \\ \Rightarrow \|G(s)\|_\infty &\leq 2(\sigma_1 + \sigma_2 + \dots + \sigma_N). \end{aligned}$$

For the case $p \neq m$ just augment $G(s)$ by zero rows or columns to make it square, but will have the same $\mathcal{L}_\infty$ norm, then the above argument gives upper bounds on the $\mathcal{L}_\infty$ norm of this augmented system. $\square$

Theorem 8.13 *Given a stable, rational, $m \times m$, transfer function $G(s)$ with Hankel singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k > \sigma_{k+1} = \sigma_{k+2} = \dots = \sigma_{k+r} > \sigma_{k+r+1} \geq \dots \geq \sigma_n > 0$*

and let $\hat{G}(s) \in \mathcal{RH}_\infty$ of McMillan degree k be an optimal Hankel norm approximation to $G(s)$ obtained in Theorem 8.6, then there exists a D_0 such that

$$\sigma_{k+1}(G) \leq \|G - \hat{G} - D_0\|_\infty \leq \sigma_{k+1}(G) + \sum_{i=1}^{n-k-r} \sigma_{i+k+r}(G).$$

Proof. The theorem follows from Corollary 8.9 and Corollary 8.12. $\square$

It should be noted that if $\hat{G}$ is an optimal Hankel norm approximation of G then $\hat{G} + D$ for any constant matrix D is also an optimal Hankel norm approximation. Hence the constant term of $\hat{G}$ can not be determined from Hankel norm.

An appropriate constant term D_0 in Theorem 8.13 can be obtained in many ways. We shall mention three of them:

- Apply Corollary 8.12 to $G(s) - \hat{G}(s)$. This is usually complicated since $(G(s) - \hat{G}(s))$ in general has McMillan degree of $n + k$.
- An alternative is to use the unstable part of the optimal Hankel norm approximation in Theorem 8.6. Let $\hat{G} + F$ be obtained from Theorem 8.6, part (3) such that $F(s) \in \mathcal{RH}_\infty^-$ has McMillan degree $\leq n - k - r$ then

$$\|G - \hat{G} - D_0\|_\infty \leq \|G - \hat{G} - F\|_\infty + \|F - D_0\|_\infty = \sigma_{k+1}(G) + \|F - D_0\|_\infty$$

Now Corollary 8.12 can be applied to $F(-s)$ to obtain a D_0 such that

$$\|F - D_0\|_\infty \leq \sum_{i=1}^{n-k-r} \sigma_i(F(-s)) \leq \sum_{i=1}^{n-k-r} \sigma_{i+k+r}(G).$$

since by Corollary 8.9,

$$\sigma_i(F(-s)) \leq \sigma_{i+k+r}(G)$$

for $i = 1, 2, \dots, n - k - r$.

- D_0 can of course be obtained using any standard convex optimization algorithm:

$$D_0 = \arg \min_{D_0} \|G - \hat{G} - D_0\|_\infty.$$

Note that Theorem 8.13 can also be applied to non-square systems. For example, consider a stable, rational, $p \times m$ ($p < m$), transfer function $G(s)$. Let $G_a = \begin{bmatrix} G \\ 0 \end{bmatrix}$ be an augmented square transfer function and let $\hat{G}_a = \begin{bmatrix} \hat{G} \\ \hat{G}_2 \end{bmatrix}$ be the optimal Hankel norm approximation of G_a such that

$$\sigma_{k+1}(G_a) \leq \|G_a - \hat{G}_a - D_a\|_\infty \leq \sigma_{k+1}(G_a) + \sum_{i=1}^{n-k-r} \sigma_{i+k+r}(G_a)$$

with $D_a = \begin{bmatrix} D_0 \\ D_2 \end{bmatrix}$. Then

$$\sigma_{k+1}(G) \leq \|G - \hat{G} - D_0\|_\infty \leq \sigma_{k+1}(G) + \sum_{i=1}^{n-k-r} \sigma_{i+k+r}(G)$$

since $\sigma_i(G) = \sigma_i(G_a)$ and $\|G - \hat{G} - D_0\|_\infty \leq \|G_a - \hat{G}_a - D_a\|_\infty$.

A less tighter error bound can be obtained without computing the appropriate D_0 .

Corollary 8.14 *Given a stable, rational, $m \times m$, strictly proper transfer function $G(s)$, with Hankel singular values $\sigma_1 \geq \sigma_2 \dots \geq \sigma_k > \sigma_{k+1} = \sigma_{k+2} \dots = \sigma_{k+r} > \sigma_{k+r+1} \geq \dots \geq \sigma_n > 0$ and let $\hat{G}(s) \in \mathcal{RH}_\infty$ of McMillan degree k be a strictly proper optimal Hankel norm approximation to $G(s)$ obtained in Theorem 8.6, then*

$$\sigma_{k+1}(G) \leq \|G - \hat{G}\|_\infty \leq 2(\sigma_{k+1}(G) + \sum_{i=1}^{n-k-r} \sigma_{i+k+r}(G))$$

Proof. The result follows from Theorem 8.13. $\square$

8.5 Bounds for Balanced Truncation

Very similar techniques to those of Theorem 8.11 can be used to bound the error obtained by truncating a balanced realization. We will first need a lemma that gives a perhaps surprising relationship between a truncated balanced realization of degree $(n - r_N)$ and an optimal Hankel norm approximation of the same degree.

Lemma 8.15 *Let (A, B, C) be a balanced realization of the stable, rational, $m \times m$ transfer function $G(s)$, and let*

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad C = \begin{bmatrix} C_1 & C_2 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \sigma I \end{bmatrix}$$

Let $(\hat{A}, \hat{B}, \hat{C}, \hat{D})$ be defined by equations (8.27) to (8.32) (where $\Sigma_2 = \Sigma_1$) and define

$$G_b := \left[\begin{array}{c|c} \frac{A_{11}}{C_1} & \frac{B_1}{0} \end{array} \right]$$

$$G_h := \left[\begin{array}{c|c} \frac{\hat{A}}{\hat{C}} & \frac{\hat{B}}{\hat{D}} \end{array} \right]$$

then

- (1) $(G_b(s) - G_h(s))/\sigma$ is all-pass.
 (2) $\|G(s) - G_b(s)\|_\infty \leq 2\sigma$.
 (3) If $\Sigma_1 > \sigma I$ then $\|G(s) - G_b(s)\|_H \leq 2\sigma$.

Proof. (1) In order to prove that $(G_b(s) - G_h(s))/\sigma$ is all-pass we note that

$$G_b(s) - G_h(s) = \left[\begin{array}{c|c} \tilde{A} & \tilde{B} \\ \hline \tilde{C} & \tilde{D} \end{array} \right]$$

where

$$\tilde{A} = \begin{bmatrix} A_{11} & 0 \\ 0 & \hat{A} \end{bmatrix}, \quad \tilde{B} = \begin{bmatrix} B_1 \\ \hat{B} \end{bmatrix}, \quad \tilde{C} = [C_1 \quad -\hat{C}], \quad \tilde{D} = -\hat{D}.$$

Now Theorem 8.4, part (1) gives that the solutions to Lyapunov equations

$$\tilde{A}\tilde{P} + \tilde{P}\tilde{A}^* + \tilde{B}\tilde{B}^* = 0 \quad (8.72)$$

$$\tilde{A}^*\tilde{Q} + \tilde{Q}\tilde{A} + \tilde{C}^*\tilde{C} = 0 \quad (8.73)$$

are

$$\tilde{P} = \begin{bmatrix} \Sigma_1 & I \\ I & \Sigma_1 \Gamma^{-1} \end{bmatrix}, \quad \tilde{Q} = \begin{bmatrix} \Sigma_1 & -\Gamma \\ -\Gamma & \Sigma_1 \Gamma \end{bmatrix} \quad (8.74)$$

(This is verified by noting that the blocks of equations (8.72) and (8.73) are also blocks of equations (8.34) and (8.35) for P_e and Q_e .) Hence $\tilde{P}\tilde{Q} = \sigma^2 I$ and by Theorem 8.3 there exists $\tilde{D}$ such that $(G_b(s) - G_h(s))/\sigma$ is all-pass. That $\tilde{D} = -\hat{D}$ is an appropriate choice is verified from equations (8.54) to (8.56) and Theorem 8.3, part (2).

(2) $(G(s) - G_b(s))/\sigma = (G(s) - G_h(s))/\sigma + (G_b(s) - G_h(s))/\sigma$ but the first term on the right hand side is all-pass by Theorem 8.4, part (2) and the second term is all-pass by part (1) above. Hence $\|G(s) - G_b(s)\|_\infty \leq 2\sigma$.

(3) Similarly using the fact that all-pass functions have unity Hankel norms gives that

$$\|G(s) - G_b(s)\|_H \leq \|G(s) - G_h(s)\|_H + \|G_b(s) - G_h(s)\|_H = 2\sigma$$

(Note that $G_h(s)$ is stable if $\Sigma_1 > \sigma I$.) □

Given the results of Lemma 8.15 bounds on the error in a truncated balanced realization are easily proved as follows.

Theorem 8.16 *Let $G(s)$ be a stable, rational, $p \times m$, transfer function with Hankel singular values $\sigma_1 > \sigma_2 > \dots > \sigma_N$, where each σ_i has multiplicity r_i and let $\tilde{G}_k(s)$ be obtained by truncating the balanced realization of $G(s)$ to the first $(r_1 + r_2 + \dots + r_k)$ states. Then*

$$(1) \quad \left\| G(s) - \tilde{G}_k(s) \right\|_\infty \leq 2(\sigma_{k+1} + \sigma_{k+2} + \dots + \sigma_N).$$

$$(2) \quad \left\| G(s) - \tilde{G}_k(s) \right\|_H \leq 2(\sigma_{k+1} + \sigma_{k+2} + \dots + \sigma_N).$$

Proof. If $p \neq m$ then augmenting B or C by zero columns or rows, respectively, will still give a balanced realization and the same argument is valid. Hence assume $p = m$. Notice that since truncation of balanced realization are also balanced, satisfying the truncated Lyapunov equations, the Hankel singular values of $\tilde{G}_i(s)$ will be $\sigma_1, \sigma_2, \dots, \sigma_i$ with multiplicities $r_1, r_2, \dots, r_i$, respectively. Also $\tilde{G}_i(s)$ can be obtained by truncating the balanced realization of $\tilde{G}_{i+1}(s)$ and hence $\|\tilde{G}_{i+1}(s) - \tilde{G}_i(s)\| \leq 2\sigma_{i+1}$ for both $\mathcal{L}_\infty$ and Hankel norms. Hence $(G_N(s) := G(s))$

$$\|G(s) - \tilde{G}_k(s)\| = \left\| \sum_{i=k}^{N-1} (\tilde{G}_{i+1}(s) - \tilde{G}_i(s)) \right\| \leq 2(\sigma_{k+1} + \sigma_{k+2} + \dots + \sigma_N)$$

for both norms, and the proof is complete. $\square$

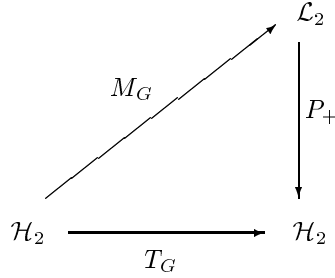
8.6 Toeplitz Operators

In this section, we consider another operator. Again let $G(s) \in \mathcal{L}_\infty$. Then a *Toeplitz operator* associated with G is denoted by T_G and is defined as

$$T_G : \mathcal{H}_2 \mapsto \mathcal{H}_2$$

$$T_G f := (P_+ M_G) f = P_+(Gf), \quad \text{for } f \in \mathcal{H}_2$$

i.e., $T_G = P_+ M_G|_{\mathcal{H}_2}$. In particular if $G(s) \in \mathcal{H}_\infty$ then $T_G = M_G|_{\mathcal{H}_2}$.



Analogous to the Hankel operator, there is also a corresponding time domain description for the Toeplitz operator:

$$T_g : \mathcal{L}_2[0, \infty) \mapsto \mathcal{L}_2[0, \infty)$$

$$T_g f := P_+(g * f) = \int_0^\infty g(t - \tau) f(\tau) d\tau, \quad t \geq 0$$

for $f(t) \in \mathcal{L}_2[0, \infty)$.

It is seen that the multiplication operator (in frequency domain) or the convolution operator (in time domain) plays an important role in the development of the Hankel and Toeplitz operators. In fact, a multiplication operator can be decomposed as several Hankel and Toeplitz operators: let $\mathcal{L}_2 = \mathcal{H}_2^\perp \oplus \mathcal{H}_2$ and $G \in \mathcal{L}_\infty$. Then the multiplication operator associated with G can be written as

$$M_G : \mathcal{H}_2^\perp \oplus \mathcal{H}_2 \mapsto \mathcal{H}_2^\perp \oplus \mathcal{H}_2$$

$$M_G = \begin{bmatrix} P_- M_G|_{\mathcal{H}_2^\perp} & P_- M_G|_{\mathcal{H}_2} \\ P_+ M_G|_{\mathcal{H}_2^\perp} & P_+ M_G|_{\mathcal{H}_2} \end{bmatrix} = \begin{bmatrix} P_- M_G|_{\mathcal{H}_2^\perp} & \Gamma_{G^\sim}^* \\ \Gamma_G & T_G \end{bmatrix}.$$

Note that $P_- M_G|_{\mathcal{H}_2^\perp}$ is actually a Toeplitz operator; however, we will not discuss it further here. A fact worth mentioning is that if G is causal, i.e., $G \in \mathcal{H}_\infty$, then $\Gamma_{G^\sim}^* = 0$ and M_G is a lower block triangular operator. In fact, it can be shown that G is causal if and only if $\Gamma_{G^\sim}^* = 0$, i.e., the future input does not affect the past output. This is yet another characterization of causality.

8.7 Hankel and Toeplitz Operators on the Disk*

It is sometimes more convenient and insightful to study operators on the disk. From the control system point of view, some operators are much easier to interpret and compute in discrete time. The objective of this section is to examine the Hankel and Toeplitz operators in discrete time (i.e., on the disk) and, hopefully, to give the readers some intuitive ideas about these operators since they are very important in the $\mathcal{H}_\infty$ control theory. To start with, it is necessary to introduce some function spaces in respect to a unit disk.

Let $\mathbb{D}$ denote the unit disk :

$$\mathbb{D} := \{\lambda \in \mathbb{C} : |\lambda| < 1\},$$

and let $\overline{\mathbb{D}}$ and $\partial\mathbb{D}$ denote its closure and boundary, respectively:

$$\overline{\mathbb{D}} := \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}$$

$$\partial\mathbb{D} := \{\lambda \in \mathbb{C} : |\lambda| = 1\}.$$

Let $\mathcal{L}_2(\partial\mathbb{D})$ denote the Hilbert space of matrix valued functions defined on the unit circle $\partial\mathbb{D}$ as

$$\mathcal{L}_2(\partial\mathbb{D}) = \left\{ F(\lambda) : \frac{1}{2\pi} \int_0^{2\pi} \text{Trace} [F^*(e^{j\theta}) F(e^{j\theta})] d\theta < \infty \right\}$$

with inner product

$$\langle F, G \rangle := \frac{1}{2\pi} \int_0^{2\pi} \text{Trace} [F^*(e^{j\theta}) G(e^{j\theta})] d\theta.$$

Furthermore, let $\mathcal{H}_2(\partial\mathbb{D})$ be the (closed) subspace of $\mathcal{L}_2(\partial\mathbb{D})$ with matrix functions $F(\lambda)$ analytic in $\mathbb{D}$, i.e.,

$$\mathcal{H}_2(\partial\mathbb{D}) = \left\{ F(\lambda) \in \mathcal{L}_2(\partial\mathbb{D}) : \frac{1}{2\pi} \int_0^{2\pi} F(e^{j\theta}) e^{jn\theta} d\theta = 0, \text{ for all } n > 0 \right\},$$

and let $\mathcal{H}_2^\perp(\partial\mathbb{D})$ be the (closed) subspace of $\mathcal{L}_2(\partial\mathbb{D})$ with matrix functions $F(\lambda)$ analytic in $\mathbb{C} \setminus \overline{\mathbb{D}}$.

It can be shown that the Fourier transform (or bilateral $\mathcal{Z}$ transform) gives the following isometric isomorphism:

$$\begin{aligned} \mathcal{L}_2(\partial\mathbb{D}) &\cong l_2(-\infty, \infty) \\ \mathcal{H}_2(\partial\mathbb{D}) &\cong l_2[0, \infty) \\ \mathcal{H}_2^\perp(\partial\mathbb{D}) &\cong l_2(-\infty, 0). \end{aligned}$$

Remark 8.3 It should be kept in mind that, in contrast to the variable z in the standard Z -transform, λ here denotes $\lambda = z^{-1}$. ♥

Analogous to the space in the half plane, $\mathcal{L}_\infty(\partial\mathbb{D})$ is used to denote the Banach space of essentially bounded matrix functions with norm

$$\|F\|_\infty = \operatorname{ess\,sup}_{\theta \in [0, 2\pi]} \overline{\sigma}[F(e^{j\theta})].$$

The Hardy space $\mathcal{H}_\infty(\partial\mathbb{D})$ is the closed subspace of $\mathcal{L}_\infty(\partial\mathbb{D})$ with functions analytic in $\mathbb{D}$ and is defined as

$$\mathcal{H}_\infty(\partial\mathbb{D}) = \left\{ F(\lambda) \in \mathcal{L}_\infty(\partial\mathbb{D}) : \int_0^{2\pi} F(e^{j\theta}) e^{jn\theta} d\theta = 0, \text{ for all } n > 0 \right\}.$$

The $\mathcal{H}_\infty$ norm is defined as

$$\|F\|_\infty := \sup_{\lambda \in \mathbb{D}} \overline{\sigma}[F(\lambda)] = \operatorname{ess\,sup}_{\theta \in [0, 2\pi]} \overline{\sigma}[F(e^{j\theta})].$$

It is easy to see that $\mathcal{L}_\infty(\partial\mathbb{D}) \subset \mathcal{L}_2(\partial\mathbb{D})$ and $\mathcal{H}_\infty(\partial\mathbb{D}) \subset \mathcal{H}_2(\partial\mathbb{D})$. (However, it should be pointed out that these inclusions are not true for functions in the half planes or for continuous time functions.)

Example 8.4 Let $F(\lambda) \in \mathcal{L}_2(\partial\mathbb{D})$, and let $F(\lambda)$ have a power series representation as follows:

$$F(\lambda) = \sum_{i=-\infty}^{\infty} F_i \lambda^i.$$

Then $F(\lambda) \in \mathcal{H}_2(\partial\mathbb{D})$ if and only if $F_i = 0$ for all $i < 0$ and $F(\lambda) \in \mathcal{H}_2^\perp(\partial\mathbb{D})$ if and only if $F_i = 0$ for all $i \geq 0$. ◇

Now let P_- and P_+ denote the orthogonal projections:

$$P_+ : l_2(-\infty, \infty) \mapsto l_2[0, \infty) \text{ or } \mathcal{L}_2(\partial\mathbb{D}) \mapsto \mathcal{H}_2(\partial\mathbb{D})$$

$$P_- : l_2(-\infty, \infty) \mapsto l_2(-\infty, 0) \text{ or } \mathcal{L}_2(\partial\mathbb{D}) \mapsto \mathcal{H}_2^\perp(\partial\mathbb{D}).$$

Suppose $G_d(\lambda) \in \mathcal{L}_\infty(\partial\mathbb{D})$. Then the Hankel operator associated with $G_d(\lambda)$ is defined as

$$\Gamma_{G_d} : l_2(-\infty, 0) \mapsto l_2[0, \infty) \text{ or } \mathcal{H}_2^\perp(\partial\mathbb{D}) \mapsto \mathcal{H}_2(\partial\mathbb{D})$$

$$\Gamma_{G_d} = P_+ M_{G_d}|_{\mathcal{H}_2^\perp(\partial\mathbb{D})}.$$

Similarly, a Toeplitz operator associated with $G_d(\lambda)$ is defined as

$$T_{G_d} : l_2[0, \infty) \mapsto l_2[0, \infty) \text{ or } \mathcal{H}_2(\partial\mathbb{D}) \mapsto \mathcal{H}_2(\partial\mathbb{D})$$

$$T_{G_d} = P_+ M_{G_d}|_{\mathcal{H}_2(\partial\mathbb{D})}.$$

Now let $G_d(\lambda) = \sum_{i=-\infty}^{\infty} G_i \lambda^i \in \mathcal{L}_\infty(\partial\mathbb{D})$ be a linear system transfer matrix, $u(\lambda) = \sum_{i=-\infty}^{\infty} u_i \lambda^i \in \mathcal{L}_2(\partial\mathbb{D})$ be the system input in the frequency domain, and u_i be the input at time $t = i$. Then the system output in the frequency domain is given by

$$y(\lambda) = G_d(\lambda)u(\lambda) = \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} G_i u_j \lambda^{i+j} =: \sum_{i=-\infty}^{\infty} y_i \lambda^i$$

where y_i is the system time response at time $t = i$, and consequently, we have

$$\begin{bmatrix} \bullet \\ \bullet \\ y_2 \\ y_1 \\ y_0 \\ y_{-1} \\ y_{-2} \\ y_{-3} \\ \bullet \\ \bullet \end{bmatrix} = \begin{bmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & G_0 & G_1 & G_2 & G_3 & G_4 & G_5 & \bullet & \bullet & \bullet \\ \bullet & \bullet & G_{-1} & G_0 & G_1 & G_2 & G_3 & G_4 & \bullet & \bullet & \bullet \\ \bullet & \bullet & G_{-2} & G_{-1} & G_0 & G_1 & G_2 & G_3 & \bullet & \bullet & \bullet \\ \bullet & \bullet & G_{-3} & G_{-2} & G_{-1} & G_0 & G_1 & G_2 & \bullet & \bullet & \bullet \\ \bullet & \bullet & G_{-4} & G_{-3} & G_{-2} & G_{-1} & G_0 & G_1 & \bullet & \bullet & \bullet \\ \bullet & \bullet & G_{-5} & G_{-4} & G_{-3} & G_{-2} & G_{-1} & G_0 & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \end{bmatrix} \begin{bmatrix} \bullet \\ \bullet \\ u_2 \\ u_1 \\ u_0 \\ u_{-1} \\ u_{-2} \\ u_{-3} \\ \bullet \\ \bullet \end{bmatrix}. \quad (8.75)$$

Equation (8.75) can be rewritten as

$$\begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \bullet \\ \bullet \\ y_{-1} \\ y_{-2} \\ y_{-3} \\ \bullet \\ \bullet \end{bmatrix} = \begin{bmatrix} G_0 & G_{-1} & G_{-2} & \bullet & \bullet & G_1 & G_2 & G_3 & \bullet & \bullet & \bullet \\ G_1 & G_0 & G_{-1} & \bullet & \bullet & G_2 & G_3 & G_4 & \bullet & \bullet & \bullet \\ G_2 & G_1 & G_0 & \bullet & \bullet & G_3 & G_4 & G_5 & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ G_{-1} & G_{-2} & G_{-3} & \bullet & \bullet & G_0 & G_1 & G_2 & \bullet & \bullet & \bullet \\ G_{-2} & G_{-3} & G_{-4} & \bullet & \bullet & G_{-1} & G_0 & G_1 & \bullet & \bullet & \bullet \\ G_{-3} & G_{-4} & G_{-5} & \bullet & \bullet & G_{-2} & G_{-1} & G_0 & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ \bullet \\ \bullet \\ u_{-1} \\ u_{-2} \\ u_{-3} \\ \bullet \\ \bullet \end{bmatrix}$$

$$=: \left[\begin{array}{c|c} T_1 & H_1 \\ \hline H_2 & T_2 \end{array} \right] \begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ \bullet \\ \bullet \\ u_{-1} \\ u_{-2} \\ u_{-3} \\ \bullet \\ \bullet \end{bmatrix}.$$

Matrices like T_1 and T_2 are called (block) *Toeplitz matrices* and matrices like H_1 and H_2 are called (block) *Hankel matrices*. In fact, H_1 is the matrix representation of the Hankel operator Γ_{G_d} and T_1 is the matrix representation of the Toeplitz operator T_{G_d} , and so on. Thus these operator norms can be computed from the matrix norms of their corresponding matrix representations.

Lemma 8.17 $\|G_d(\lambda)\|_\infty = \left\| \begin{bmatrix} T_1 & H_1 \\ H_2 & T_2 \end{bmatrix} \right\|$, $\|\Gamma_{G_d}\| = \|H_1\|$, and $\|T_{G_d}\| = \|T_1\|$.

Example 8.5 Let $G_d(\lambda) \in \mathcal{RH}_\infty(\partial\mathbb{D})$ and $G_d(\lambda) = C(\lambda^{-1}I - A)^{-1}B + D$ be a state space realization of $G_d(\lambda)$ and let A have all the eigenvalues in $\mathbb{D}$. Then

$$G_d(\lambda) = D + \sum_{i=0}^{\infty} \lambda^{i+1} C A^i B = \sum_{i=0}^{\infty} G_i \lambda^i$$

with $G_0 = D$ and $G_i = C A^{i-1} B, \forall i \geq 1$. The state space equations are given by

$$\begin{aligned} x_{k+1} &= A x_k + B u_k, \quad x_{-\infty} = 0, \\ y_k &= C x_k + D u_k. \end{aligned}$$

Then for $u \in l_2(-\infty, 0)$, we have $x_0 = \sum_{i=1}^{\infty} A^{i-1} B u_{-i}$, which defines the controllability operator

$$x_0 = \Psi_c u = \begin{bmatrix} B & AB & A^2 B & \cdots \end{bmatrix} \begin{bmatrix} u_{-1} \\ u_{-2} \\ u_{-3} \\ \vdots \end{bmatrix} \in \mathbb{R}^n.$$

On the other hand, given x_0 and $u_k = 0, i \geq 0, x \in \mathbb{R}^n$, the output can be computed as

$$\Psi_o x_0 = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \end{bmatrix} x_0 \in l_2[0, \infty)$$

which defines the observability operator. Of course, the adjoint operators of Ψ_c and Ψ_o can be computed easily as

$$\Psi_c^* x_0 = \begin{bmatrix} B^* \\ B^* A^* \\ B^* (A^*)^2 \\ \vdots \end{bmatrix} x_0 \in l_2(-\infty, 0)$$

and

$$\Psi_o^* y = \begin{bmatrix} C^* & A^* C^* & (A^*)^2 C^* & \cdots \end{bmatrix} \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \end{bmatrix} \in \mathbb{R}^n.$$

Hence, the Hankel operator has the following matrix representation:

$$H = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \end{bmatrix} \begin{bmatrix} B & AB & A^2 B & \cdots \end{bmatrix} = \begin{bmatrix} G_1 & G_2 & G_3 & \cdots \\ G_2 & G_3 & G_4 & \cdots \\ G_3 & G_4 & G_5 & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}.$$

◇

It is interesting to establish some connections between a Hankel operator defined on the unit disk and the one defined on the half plane. First define the map as

$$\lambda = \frac{s-1}{s+1}, \quad s = \frac{1+\lambda}{1-\lambda}, \quad (8.76)$$

which maps the closed right-half plane $\operatorname{Re}(s) \geq 0$ onto the closed unit disk, $\overline{\mathbb{D}}$.

Suppose $G(s) \in \mathcal{H}_\infty(j\mathbb{R})$ and define

$$G_d(\lambda) := G(s)|_{s=\frac{1+\lambda}{1-\lambda}}. \quad (8.77)$$

Since $G(s)$ is analytic in $\operatorname{Re}(s) > 0$, including the point at ∞ , $G_d(\lambda)$ is analytic in $\mathbb{D}$, i.e., $G_d(\lambda) \in \mathcal{H}_\infty(\partial\mathbb{D})$.

Lemma 8.18 $\|\Gamma_{G(s)}\| = \|\Gamma_{G_d(\lambda)}\|$.

Proof. Define the function

$$\phi(s) = \frac{\sqrt{2}}{s+1}.$$

The relation between a point $j\omega$ on the imaginary axis and the corresponding point $e^{j\theta}$ on the unit circle is, from (8.76),

$$e^{j\theta} = \frac{j\omega - 1}{j\omega + 1}.$$

This yields

$$d\theta = -\frac{2}{\omega^2 + 1}d\omega = -|\phi(j\omega)|^2 d\omega,$$

which implies that the mapping

$$f_d(\lambda) \mapsto \phi(s)f(s) : \mathcal{H}_2(\partial\mathbb{D}) \mapsto \mathcal{H}_2(j\mathbb{R})$$

where $f(s) = f_d(\lambda)|_{\lambda=\frac{s-1}{s+1}}$ is an isomorphism. Similarly,

$$f_d(\lambda) \mapsto \phi(s)f(s) : \mathcal{H}_2^\perp(\partial\mathbb{D}) \mapsto \mathcal{H}_2^\perp(j\mathbb{R})$$

is an isomorphism; note that if $f_d(\lambda) \in \mathcal{H}_2^\perp(\partial\mathbb{D})$, then $f_d(\infty) = 0$, so that $f(-1) = 0$, and hence $\phi(s)f(s)$ is analytic in $\operatorname{Re}(s) < 0$.

The lemma now follows from the commutative diagram

$$\begin{array}{ccc} \mathcal{H}_2^\perp(\partial\mathbb{D}) & \xrightarrow{\Gamma_{G_d(\lambda)}} & \mathcal{H}_2(\partial\mathbb{D}) \\ \downarrow & & \downarrow \\ \mathcal{H}_2^\perp(j\mathbb{R}) & \xrightarrow{\Gamma_{G(s)}} & \mathcal{H}_2(j\mathbb{R}) \end{array}$$

□

The above isomorphism between $\mathcal{H}_2(\partial\mathbb{D})$ and $\mathcal{H}_2(j\mathbb{R})$ has also the implication that every discrete time $\mathcal{H}_2$ control problem can be converted to an equivalent continuous time $\mathcal{H}_2$ control problem. It should be emphasized that the bilinear transformation is not an isomorphism between $\mathcal{H}_2(\partial\mathbb{D})$ and $\mathcal{H}_2(j\mathbb{R})$.

8.8 Nehari's Theorem*

As we have mentioned at the beginning of this chapter, a Hankel operator is closely related to an analytic approximation problem. In this section, we shall be concerned with approximating G by an anticausal transfer matrix, i.e., one analytic in $\operatorname{Re}(s) < 0$ (or $|\lambda| > 1$), where the approximation is with respect to the $\mathcal{L}_\infty$ norm. For convenience, let $\mathcal{H}_\infty^-(\partial\mathbb{D})$ denote the subspace of $\mathcal{L}_\infty(\partial\mathbb{D})$ with functions analytic in $|\lambda| > 1$. So $G(\lambda) \in \mathcal{H}_\infty^-(\partial\mathbb{D})$ if and only if $G(\lambda^{-1}) \in \mathcal{H}_\infty(\partial\mathbb{D})$.

Minimum Distance Problem

In the following, we shall establish that the distance in $\mathcal{L}_\infty$ from G to the nearest matrix in $\mathcal{H}_\infty^-$ equals $\|\Gamma_G\|$. This is the classical Nehari Theorem. Some applications

and explicit construction of the approximation for the matrix case will be considered in the later chapters.

Theorem 8.19 *Suppose $G \in \mathcal{L}_\infty$, then*

$$\inf_{Q \in \mathcal{H}_\infty^-} \|G - Q\|_\infty = \|\Gamma_G\|$$

and the infimum is achieved.

Remark 8.4 Note that from the mapping (8.76) and Lemma 8.18, the above theorem is the same whether the problem is on the unit disk or on the half plane. Hence, we will only give the proof on the unit disk. $\heartsuit$

Proof. We shall give the proof on the unit disk. First, it is easy to establish that the Hankel norm is the lower bound: for fixed $Q \in \mathcal{H}_\infty^-(\partial\mathbb{D})$, we have

$$\begin{aligned} \|G - Q\|_\infty &= \sup_{f \in \mathcal{H}_2^\perp(\partial\mathbb{D})} \frac{\|(G - Q)f\|_2}{\|f\|_2} \\ &\geq \sup_{f \in \mathcal{H}_2^\perp(\partial\mathbb{D})} \frac{\|P_+(G - Q)f\|_2}{\|f\|_2} \\ &= \sup_{f \in \mathcal{H}_2^\perp(\partial\mathbb{D})} \frac{\|P_+Gf\|_2}{\|f\|_2} \\ &= \|\Gamma_G\|. \end{aligned}$$

Next we shall construct a $Q(\lambda) \in \mathcal{H}_\infty^-(\partial\mathbb{D})$ such that

$$\|G - Q\|_\infty = \|\Gamma_G\|. \quad (8.78)$$

Let the power series expansion of $G(\lambda)$ and $Q(\lambda)$ be

$$\begin{aligned} G(\lambda) &= \sum_{i=-\infty}^{\infty} \lambda^i G_i \\ Q(\lambda) &= \sum_{i=-\infty}^0 \lambda^i Q_i. \end{aligned}$$

The left-hand side of (8.78) equals the norm of the operator

$$f \mapsto (G(\lambda) - Q(\lambda))f(\lambda) : \mathcal{H}_2^\perp(\partial\mathbb{D}) \mapsto \mathcal{L}_2(\partial\mathbb{D}).$$

From the discussion of the last section, there is a matrix representation of this operator:

$$\begin{bmatrix} \bullet & & \bullet & & \bullet & \bullet & \bullet \\ \bullet & & \bullet & & \bullet & \bullet & \bullet \\ G_3 & & G_4 & & G_5 & \bullet & \bullet \\ G_2 & & G_3 & & G_4 & \bullet & \bullet \\ G_1 & & G_2 & & G_3 & \bullet & \bullet \\ G_0 - Q_0 & & G_1 & & G_2 & \bullet & \bullet \\ G_{-1} - Q_{-1} & G_0 - Q_0 & & G_1 & & \bullet & \bullet \\ G_{-2} - Q_{-2} & G_{-1} - Q_{-1} & G_0 - Q_0 & & & \bullet & \bullet \\ \bullet & & \bullet & & \bullet & \bullet & \bullet \\ \bullet & & \bullet & & \bullet & \bullet & \bullet \end{bmatrix}. \quad (8.79)$$

The idea in the construction of a Q to satisfy (8.78) is to select $Q_0, Q_{-1}, \dots$, in turn to minimize the matrix norm of (8.79). First choose Q_0 to minimize

$$\left\| \left[\begin{array}{ccc|ccc} \bullet & & \bullet & \bullet & \bullet & \bullet \\ \bullet & & \bullet & \bullet & \bullet & \bullet \\ G_3 & & G_4 & G_5 & \bullet & \bullet \\ G_2 & & G_3 & G_4 & \bullet & \bullet \\ G_1 & & G_2 & G_3 & \bullet & \bullet \\ \hline G_0 - Q_0 & & G_1 & G_2 & \bullet & \bullet \end{array} \right] \right\|.$$

By Parrott's Theorem (i.e., matrix dilation theory presented in Chapter 2), the minimum equals the norm of the following matrix

$$\begin{bmatrix} \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet \\ G_3 & G_4 & G_5 & \bullet & \bullet \\ G_2 & G_3 & G_4 & \bullet & \bullet \\ G_1 & G_2 & G_3 & \bullet & \bullet \end{bmatrix}$$

which is the rotated Hankel matrix H_1 . Hence, the minimum equals the Hankel norm $\|\Gamma_G\|$. Next, choose Q_{-1} to minimize

$$\left\| \left[\begin{array}{ccc|ccc} \bullet & & \bullet & \bullet & \bullet & \bullet \\ \bullet & & \bullet & \bullet & \bullet & \bullet \\ G_3 & & G_4 & G_5 & \bullet & \bullet \\ G_2 & & G_3 & G_4 & \bullet & \bullet \\ G_1 & & G_2 & G_3 & \bullet & \bullet \\ \hline G_0 - Q_0 & & G_1 & G_2 & \bullet & \bullet \\ \hline G_{-1} - Q_{-1} & & G_0 - Q_0 & G_1 & \bullet & \bullet \end{array} \right] \right\|.$$

Again, the minimum equals $\|\Gamma_G\|$. Continuing in this way gives a suitable Q . $\square$

As we have mentioned earlier we might be interested in approximating an $\mathcal{L}_\infty$ function by an $\mathcal{H}_\infty$ function. Then we have the following corollary.

Corollary 8.20 Suppose $G \in \mathcal{L}_\infty$, then

$$\inf_{Q \in \mathcal{H}_\infty} \|G - Q\|_\infty = \|\hat{\Gamma}_G\|$$

and the infimum is achieved.

Proof. This follows from the fact that

$$\inf_{Q \in \mathcal{H}_\infty} \|G - Q\|_\infty = \inf_{Q^\sim \in \mathcal{H}_\infty^-} \|G^\sim - Q^\sim\|_\infty = \|\Gamma_{G^\sim}\| = \|\hat{\Gamma}_G\|.$$

□

Let $G \in \mathcal{RH}_\infty$ and g be the impulse response of G and let σ_1 be the largest Hankel singular value of G , i.e., $\|\Gamma_G\| = \sigma_1$. Suppose $u \in \mathcal{L}_2(-\infty, 0]$ (or $l_2(-\infty, 0)$) and $v \in \mathcal{L}_2[0, \infty)$ (or $l_2[0, \infty)$) are the corresponding Schmidt pairs:

$$\begin{aligned} \Gamma_g u &= \sigma_1 v \\ \Gamma_g^* v &= \sigma_1 u. \end{aligned}$$

Now denote the Laplace transform (or Z -transform) of u and v as $U \in \mathcal{RH}_2^\perp$ and $V \in \mathcal{RH}_2$.

Lemma 8.21 Let $G \in \mathcal{RH}_\infty$ and σ_1 be the largest Hankel singular value of G . Then

$$\inf_{Q \in \mathcal{H}_\infty^-} \|G - Q\|_\infty = \sigma_1$$

and

$$(G - Q)U = \sigma_1 V.$$

Moreover, if G is a scalar function, then $Q = G - \sigma_1 V/U$ is the unique solution and $G - Q$ is all-pass.

Proof. Let $H := (G - Q)U$ and note that $\Gamma_G U \in \mathcal{RH}_2$ and $P_+ H = P_+(GU) = \Gamma_G U$. Then

$$\begin{aligned} 0 &\leq \|H - \Gamma_G U\|_2^2 \\ &= \|H\|_2^2 + \|\Gamma_G U\|_2^2 - \langle H, \Gamma_G U \rangle - \langle \Gamma_G U, H \rangle \\ &= \|H\|_2^2 + \|\Gamma_G U\|_2^2 - \langle P_+ H, \Gamma_G U \rangle - \langle \Gamma_G U, P_+ H \rangle \\ &= \|H\|_2^2 - \langle \Gamma_G U, \Gamma_G U \rangle \\ &= \|H\|_2^2 - \langle U, \Gamma_G^* \Gamma_G U \rangle \\ &= \|H\|_2^2 - \sigma_1^2 \langle U, U \rangle \\ &= \|H\|_2^2 - \sigma_1^2 \|U\|_2^2 \\ &\leq \|G - Q\|_\infty^2 \|U\|_2^2 - \sigma_1^2 \|U\|_2^2 \\ &= 0. \end{aligned}$$

Hence, $H = \Gamma_G U$, i.e., $(G - Q)U = \Gamma_G U = \sigma_1 V$.

Now if G is a scalar function, then Q is uniquely determined and $Q = G - \sigma_1 V/U$. We shall prove through explicit construction below that V/U is an all-pass. $\square$

Formulas for Continuous Time

Let $G(s) \in \mathcal{RH}_\infty$ and let

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array} \right].$$

Let L_c and L_o be the corresponding controllability and observability Gramians:

$$AL_c + L_c A^* + BB^* = 0$$

$$A^* L_o + L_o A + C^* C = 0.$$

And let σ_1^2 , η be the largest eigenvalue and the corresponding eigenvector of $L_c L_o$:

$$L_c L_o \eta = \sigma_1^2 \eta.$$

Define

$$\xi := \frac{1}{\sigma_1} L_o \eta.$$

Then

$$\begin{aligned} u &= \Psi_c^* \xi = B^* e^{-A^* \tau} \xi \in \mathcal{L}_2(-\infty, 0] \\ v &= \Psi_o \eta = C e^{At} \eta \in \mathcal{L}_2[0, \infty) \end{aligned}$$

are the Schmidt pair and

$$\begin{aligned} U(s) &= \left[\begin{array}{c|c} -A^* & \xi \\ \hline -B^* & 0 \end{array} \right] \in \mathcal{RH}_2^\perp \\ V(s) &= \left[\begin{array}{c|c} A & \eta \\ \hline C & 0 \end{array} \right] \in \mathcal{RH}_2. \end{aligned}$$

It is easy to show that if G is a scalar, then U, V are scalar transfer functions and $U \sim U = V \sim V$. Hence, V/U is an all-pass. The details are left as an exercise for the reader.

Formulas for Discrete Time

Similarly, let G be a discrete time transfer matrix and let

$$G(\lambda) = C(\lambda^{-1}I - A)^{-1}B = \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array} \right].$$

Let L_c and L_o be the corresponding controllability and observability Gramians:

$$AL_cA^* - L_c + BB^* = 0$$

$$A^*L_oA - L_o + C^*C = 0.$$

And let σ_1^2 , η be the largest eigenvalue and a corresponding eigenvector of L_cL_o :

$$L_cL_o\eta = \sigma_1^2\eta.$$

Define

$$\xi := \frac{1}{\sigma_1}L_o\eta.$$

Then

$$\begin{aligned} u &= \Psi_c^*\xi = \begin{bmatrix} B^* \\ B^*A^* \\ B^*(A^*)^2 \\ \vdots \end{bmatrix} \xi \in l_2(-\infty, 0) \\ v &= \Psi_o\eta = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \end{bmatrix} \eta \in l_2[0, \infty) \end{aligned}$$

are the Schmidt pair and

$$\begin{aligned} U(\lambda) &= \sum_{i=1}^{\infty} B^*(A^*)^{i-1}\xi\lambda^{-i} = \left[\frac{A}{\xi^*} \middle| \frac{B}{0} \right]^{\sim} \in \mathcal{RH}_2^{\perp} \\ V(\lambda) &= \sum_{i=0}^{\infty} CA^i\eta\lambda^i = \left[\frac{A}{CA} \middle| \frac{\eta}{C\eta} \right] \in \mathcal{RH}_2 \end{aligned}$$

where $W(\lambda)^{\sim} := W^T(\lambda^{-1})$.

Alternatively, since the Hankel operator has a matrix representation H , u and v can be obtained from a “singular value decomposition” (possibly infinite size): let

$$G(\lambda) = \sum_{i=1}^{\infty} G_i\lambda^i, \quad G_i \in \mathbb{R}^{p \times q}$$

(in fact, $G_i = CA^{i-1}B$ if a state space realization of G is available) and let

$$H = \begin{bmatrix} G_1 & G_2 & G_3 & \cdots \\ G_2 & G_3 & G_4 & \cdots \\ G_3 & G_4 & G_5 & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}.$$

Suppose H has a “singular value decomposition”

$$\begin{aligned} H &= U \Sigma V^* \\ \Sigma &= \text{diag}\{\sigma_1, \sigma_2, \dots\} \\ U &= \begin{bmatrix} u_1 & u_2 & \dots \end{bmatrix} \\ V &= \begin{bmatrix} v_1 & v_2 & \dots \end{bmatrix} \end{aligned}$$

with $U^*U = UU^* = I$ and $V^*V = VV^* = I$. Then

$$u = u_1 = \begin{bmatrix} u_{11} \\ u_{12} \\ \vdots \end{bmatrix}, v = v_1 = \begin{bmatrix} v_{11} \\ v_{12} \\ \vdots \end{bmatrix}$$

where u and v are partitioned such that $u_{1i} \in \mathbb{R}^p$ and $v_{1i} \in \mathbb{R}^q$. Finally, $U(\lambda)$ and $V(\lambda)$ can be obtained as

$$\begin{aligned} U(\lambda) &= \sum_{i=1}^{\infty} u_{1i} \lambda^{-i} \in \mathcal{RH}_2^\perp \\ V(\lambda) &= \sum_{i=1}^{\infty} v_{1i} \lambda^{i-1} \in \mathcal{RH}_2. \end{aligned}$$

In particular, if $G(\lambda)$ is an n -th order matrix polynomial, then matrix H has only a finite number of nonzero elements and

$$H = \begin{bmatrix} H_n & 0 \\ 0 & 0 \end{bmatrix}$$

with

$$H_n = \begin{bmatrix} G_1 & G_2 & \dots & G_{n-1} & G_n \\ G_2 & G_3 & \dots & G_n & 0 \\ G_3 & G_4 & \dots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ G_n & 0 & \dots & 0 & 0 \end{bmatrix}.$$

Hence, u and v can be obtained from a standard SVD of H_n .

8.9 Notes and References

The study of the Hankel operators and of the optimal Hankel norm approximation theory can be found in Adamjan, Arov, and Krein [1978], Bettayeb, Silverman, and Safonov [1980], Kung and Lin [1981], Power [1982], Francis [1987], Kavranoğlu and Bettayeb [1994] and references therein. The presentation of this chapter is based on Francis [1987] and Glover [1984,1989].